



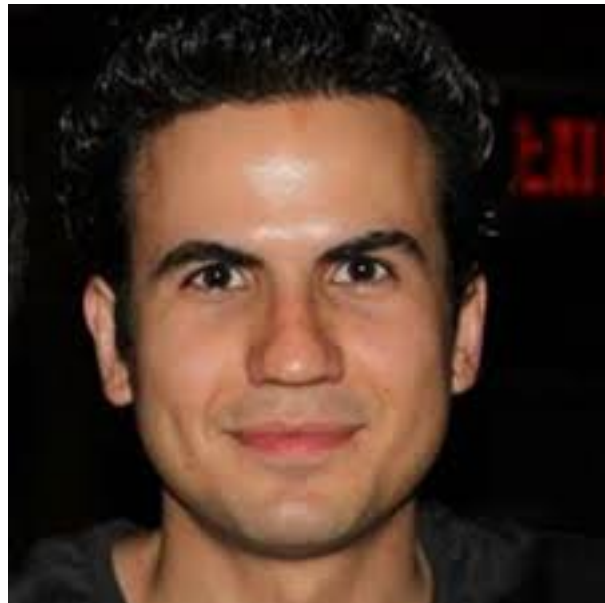
Indian Institute of
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Maxwell formulations and superconductors

Joint work with



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Integral representations for Maxwell's equations and their idiosyncrasies

	EFIE	MFIE	CFIE	ECCIE	NRCCIE	RCSIE	auRCSIE	DPIE
Resonance-free			✓		✓	✓	✓	✓
No high-density mesh breakdown		✓		✓	✓	✓	✓	✓
Free from catastrophic cancellation in $\mathbf{E}^{\text{scat}}$, $\mathbf{H}^{\text{scat}}$				✓	✓		✓	✓
Second kind Fredholm eq.		✓		✓		✓	✓	✓
Rapid convergence with GMRES, BiCGSTAB		✓		✓	✓	✓	✓	✓
Stable at low frequencies for surfaces of genus zero		✓		✓	✓	✓	✓	✓
Stable at low frequencies for surfaces of any genus								✓

Boundary conditions on a PEC surface

Tangential components

$$n \times H = J$$

$$n \times E = 0$$

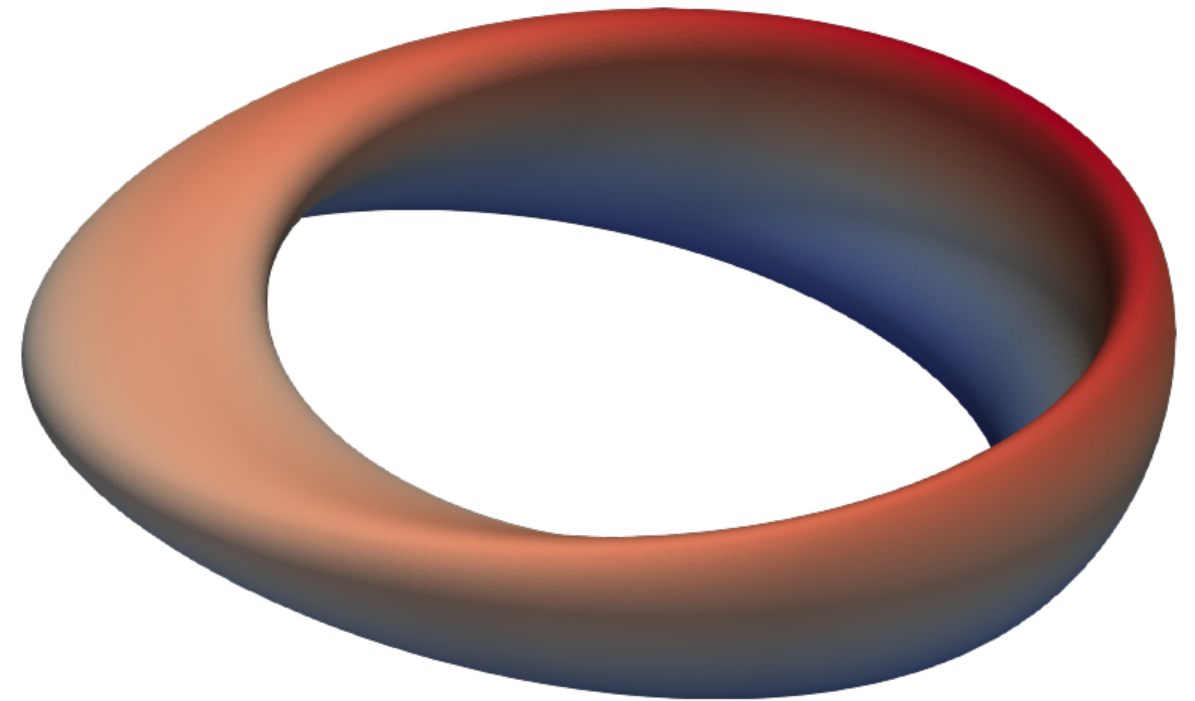
Normal components

$$n \cdot H = 0$$

$$n \cdot E = \frac{\rho}{\epsilon}$$

Continuity conditions

$$\nabla_{\Gamma} \cdot J = i\omega\rho$$



$$\nabla \times H = -i\omega\epsilon E$$

$$\nabla \times E = i\omega\mu H$$

Any two linearly independent scalar equations uniquely determine the solutions

Layer potentials and their properties

Scalar densities, scalar outputs

$$S[\rho], n \cdot \nabla S[\rho]$$

Scalar densities vector outputs

$$n \times \nabla S[\rho], n \times S[n\rho]$$

Vector densities, scalar outputs

$$n \cdot S[J], \nabla \cdot S[J]$$

Vector densities, vector outputs

$$n \times S[J], n \times \nabla \times S[J], n \times \nabla \times \nabla \times S[J]$$

$$S[\rho] = \int_{\Gamma} \frac{e^{ik|x-y|}}{4\pi|x-y|} \rho(y) dS$$

ρ : Scalar density (charges)

J : vector density (currents),
typically tangential

Compact, Fredholm, Bounded,
Hypersingular

the FIE family

Uses physical currents, and charges

$$E^s = i\omega\mu S[J] - \frac{1}{\varepsilon} \nabla S[\rho] = i\omega\mu \left(I + \frac{1}{k^2} \nabla \nabla \cdot \right) S[J]$$

$$H^s = \nabla \times S[J]$$

MFIE:

$$J/2 - n \times \nabla \times S[J] = n \times H^i$$

$$n \times H = J$$

Resonances, difficulty in evaluating E

EFIE:

$$-i\omega\mu \left(n \times n \times \left(I + \frac{1}{k^2} \nabla \nabla \cdot \right) S[J] \right) = n \times n \times E^i \quad n \times n \times E = 0$$

Hypersingular operator, low frequency breakdown

CFIE:

$$\alpha \text{ MFIE} + \beta \text{ EFIE}$$

the CCIE family

Uses physical currents, and charges

$$E^s = i\omega\mu S[J] - \frac{1}{\varepsilon} \nabla S[\rho] = i\omega\mu \left(I + \frac{1}{k^2} \nabla \nabla \cdot \right) S[J]$$

$$H^s = \nabla \times S[J]$$

MFIE:

$$J/2 - n \times \nabla \times S[J] = n \times H^i$$

$$n \times H = J$$

ECCIE: Given J , solve for an auxiliary equation for

ρ

$$\rho/2 + S'[\rho] = \varepsilon n E^i + i\omega\mu \varepsilon n \cdot S[J]$$

$$\varepsilon n \cdot E = \rho$$

Resonances, difficulty in evaluating E

NRCCIE: Coupled equations for J, ρ

$$\alpha n \times n \times \left(i\omega\mu S[J] - \frac{1}{\varepsilon} \nabla S[\rho] \right) + J/2$$

$$n \times H + \alpha n \times n \times E = J$$

$$-n \times \nabla \times S[J] = n \times H^i - \alpha n \times n \times E^i$$

$$\rho/2 + S'[\rho] - i\omega\mu \varepsilon n \cdot S[J] + \alpha \left(\nabla \cdot S[J] \right.$$

$$\left. \varepsilon n \cdot E + \alpha (S[\nabla_{\Gamma} \cdot J - i\omega\rho]) \right) = \rho$$

$$\left. -i\omega S[\rho] \right) = \varepsilon n \cdot E^i$$

Resonances, difficulty in evaluating E ,
bounded operators

the CSIE family

$$E^s = \nabla \times S[M] + i\alpha \nabla \times \nabla \times S[J]$$

$$H^s = \alpha\omega\varepsilon \nabla \times S[J] + \frac{1}{i\omega\mu} \nabla \times \nabla \times S[M]$$

CSIE:

$$n \times S[M] + i\alpha n \times \nabla \times \nabla \times S[n \times M] = -n \times E^i \quad n \times E = 0$$

RCSIE: Regularize J more to fix hypersingular
 issue

$$n \times S[M] + i\alpha n \times \nabla \times \nabla \times S[n \times S_{ik}[M]] = -n \times E^i \quad n \times E = 0$$

~~Hypersingular operator~~, difficulty in evaluating H

the CSIE family

$$E^s = \nabla \times S[M] + i\alpha \nabla \times \nabla \times S[J]$$

$$H^s = \alpha\omega\varepsilon \nabla \times S[J] + i\omega\varepsilon S[M] + \nabla (S[\sigma] + i\beta D S_{i|k}[\sigma])$$

CSIE:

$$n \times S[M] + i\alpha n \times \nabla \times \nabla \times S[n \times M] = -n \times E^i \quad n \times E = 0$$

RCSIE: Regularize J more to fix hypersingular issue

$$n \times S[M] + i\alpha n \times \nabla \times \nabla \times S[n \times S_{ik}[M]] = -n \times E^i \quad n \times E = 0$$

auRCSIE: Regularize J more to fix hypersingular issue, use augmented variable to fix evaluation issue

$$n \times S[M] + i\alpha n \times \nabla \times \nabla \times S[n \times S_{ik}[M]] = -n \times E^i \quad n \times E = 0$$

$$-\sigma/2 + S'[\sigma] + i\beta D[S_{ik}[\sigma]] = -n \cdot H^i + f(M) \quad n \cdot H = 0$$

~~Hypersingular operator, difficulty in evaluating H~~

Generalized Debye

Symmetric treatment of Electric and Magnetic fields, and relying on Hodge decomposition of currents, only 2 scalar unknowns

$$E^s = ikS[J] - \nabla S[\rho] - \nabla \times S[M]$$

$$H^s = \nabla \times S[J] - \nabla \times S[\mu] + ikS[M]$$

$$J = ik(\nabla_{\Gamma} \Delta_{\Gamma}^{-1} \rho - \nabla_{\Gamma} \Delta_{\Gamma} \mu)$$

$$M = ik(\nabla_{\Gamma} \Delta_{\Gamma}^{-1} \mu + \nabla_{\Gamma} \Delta_{\Gamma} \rho)$$

Boundary conditions imposed

$$n \cdot H = 0$$

$$S_0[\nabla_{\Gamma} \cdot n \times n \times E] = 0$$

Too many FMMs, harmonic fields needed

Generalized Debye

Symmetric treatment of Electric and Magnetic fields, and relying on Hodge decomposition of currents, only 2 scalar unknowns

$$E^s = ikS[J] - \nabla S[\rho] - \nabla \times S[M]$$

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Boundary conditions imposed

$$n \cdot H = 0$$

$$S_0[\nabla_{\Gamma} \cdot n \times n \times E] = 0$$

~~Too many FMMs~~, harmonic fields needed

Surfacefun to the rescue

DPIE

Derive PDEs for the scalar and vector potentials stable in the static limit

$$E^s = i\omega A - \nabla \phi$$

$$H^s = \frac{1}{\mu} \nabla \times A$$

$$(\Delta + k^2)\phi^s = 0$$

$$\phi|_{\Gamma_j} = V_j$$

$$\int_{\Gamma_j} \partial_n \phi = 0$$

$$(\Delta + k^2)A^s = 0$$

$$n \times A|_{\Gamma_j} = 0$$

$$\nabla \cdot A|_{\Gamma_j} = v_j$$

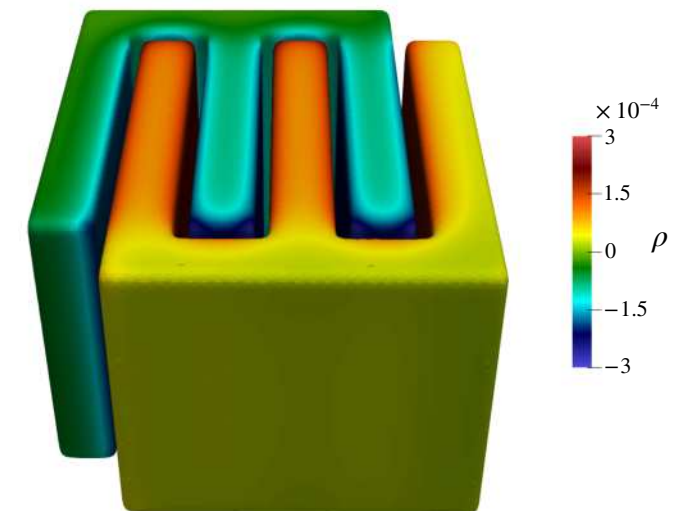
$$\int_{\Gamma_j} n \cdot A = 0$$

$$\phi^s = D[\sigma] - i\alpha S[\sigma]$$

$$A^s = \nabla \times S[a] - S[n\rho] + i\alpha(S[n \times a] + \nabla S[\rho])$$

Layer potential evaluator: fmm3dbie

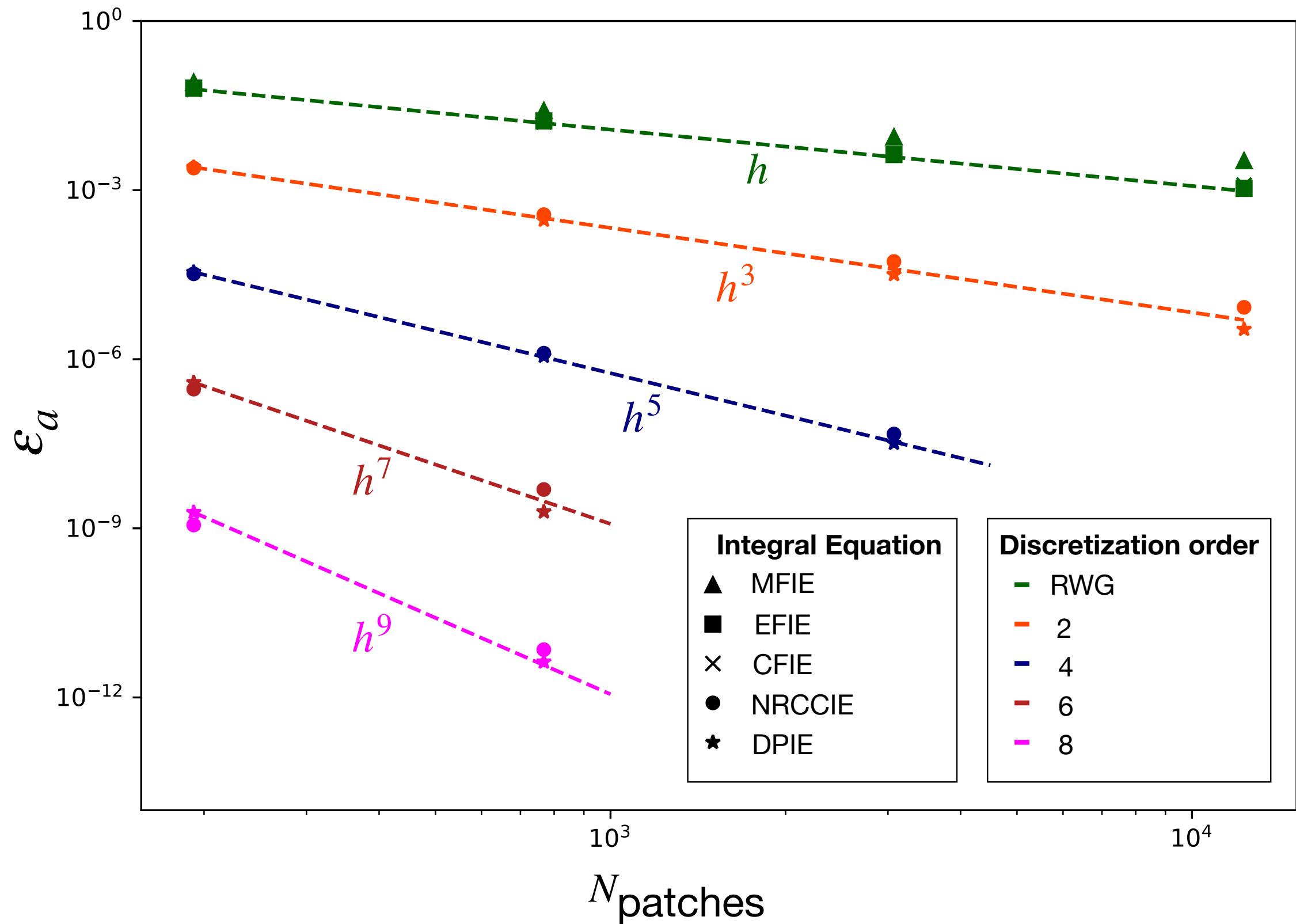
- Easy to use fortran solvers for:
 - ➔ Helmholtz equation: Dirichlet, Neumann, Impedance, Transmission
 - ➔ Laplace equation: Dirichlet
 - ➔ Stokes equation: Velocity
- Beta version solvers
 - ➔ Maxwell: PEC (NRCCIE, MFIE, CFIE, DPIE), dielectric interfaces (Muller, DFIE)
- Tutorials for building your own FMM accelerated solvers for testing new integral representations
- Standardization of geometry input format (go3) for high order solvers
- A converter for gmsh triangulations to go3 format
- Beta version of matrix entry generators for interfacing with fast direct solvers (currently undocumented on website)



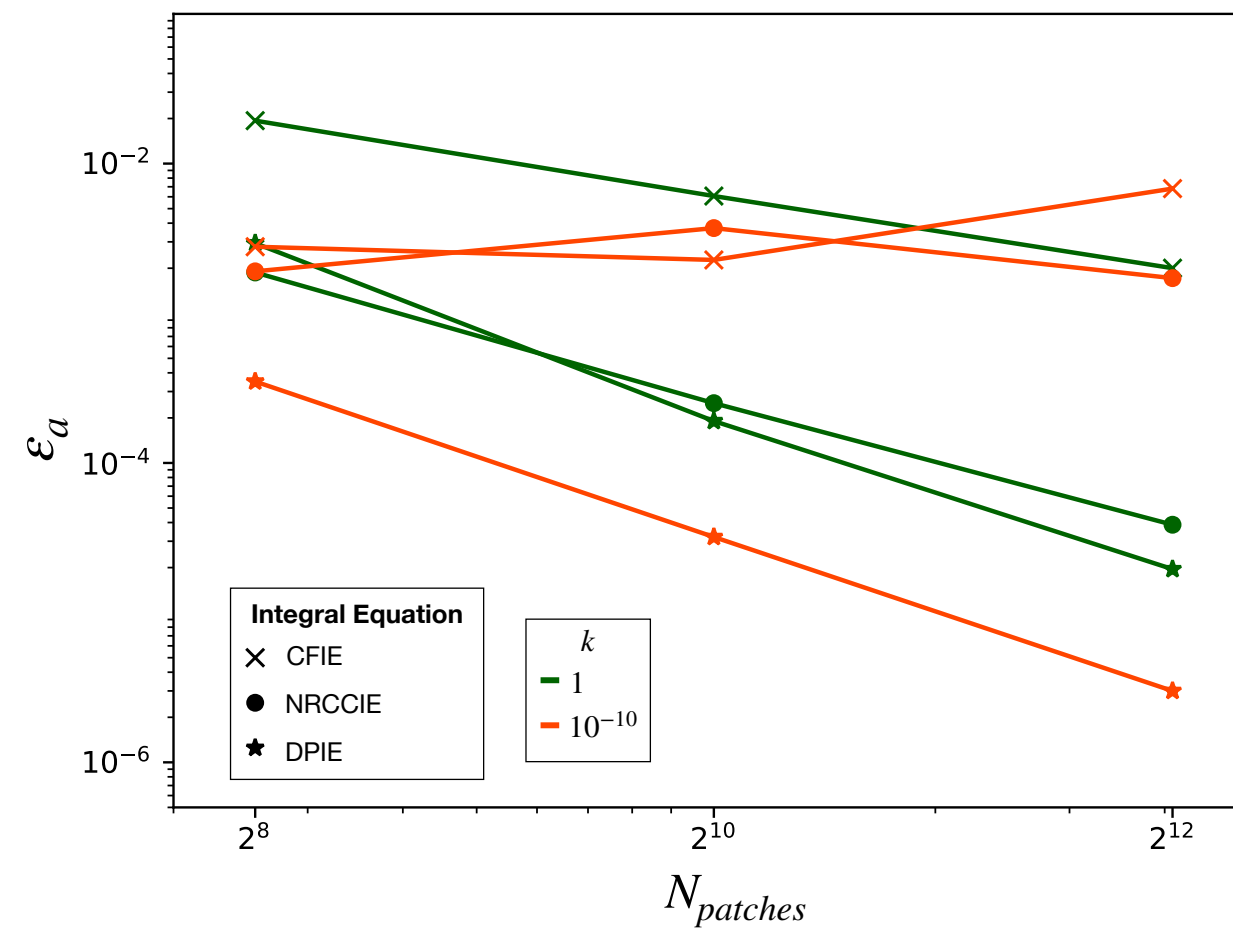
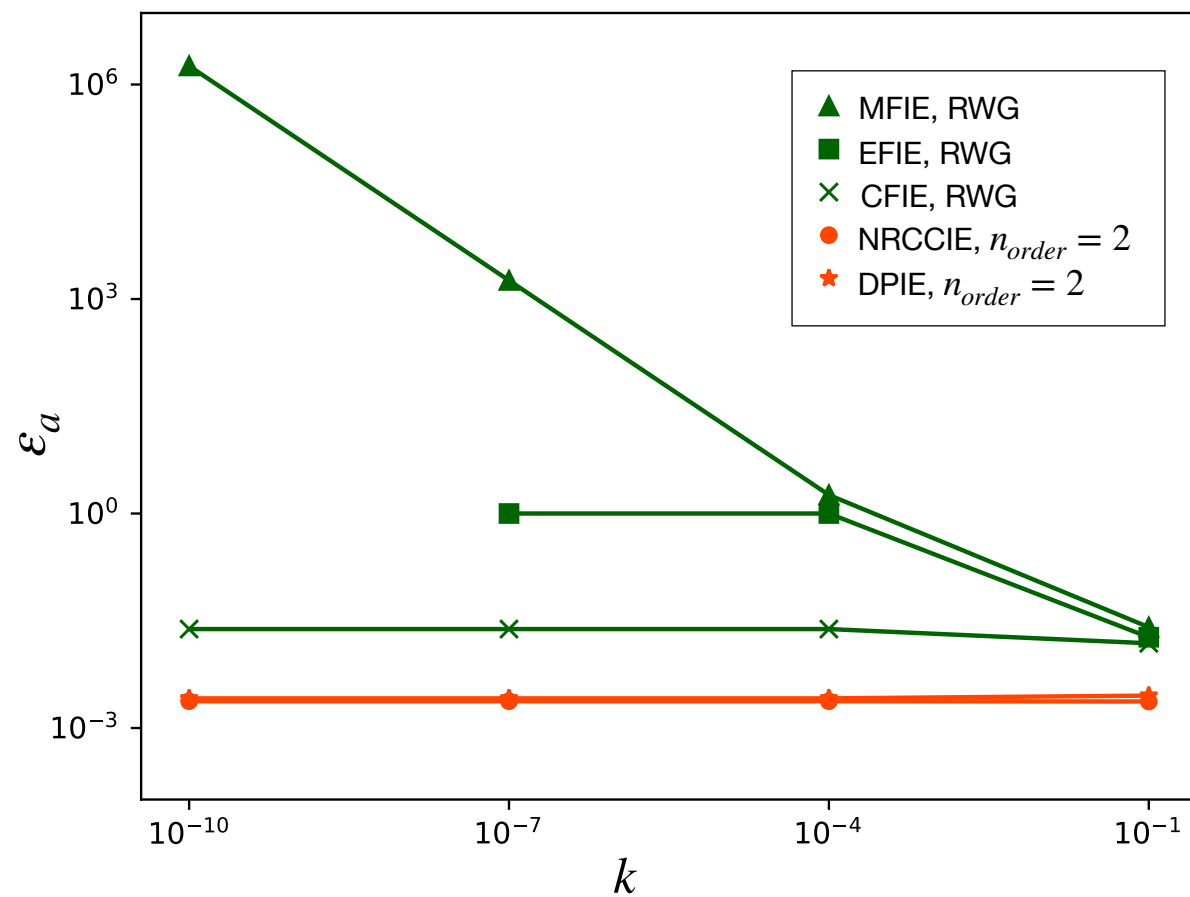
fmm3dbie.readthedocs.io/en/latest

<https://github.com/fastalgorithms/fmm3dbie.git>

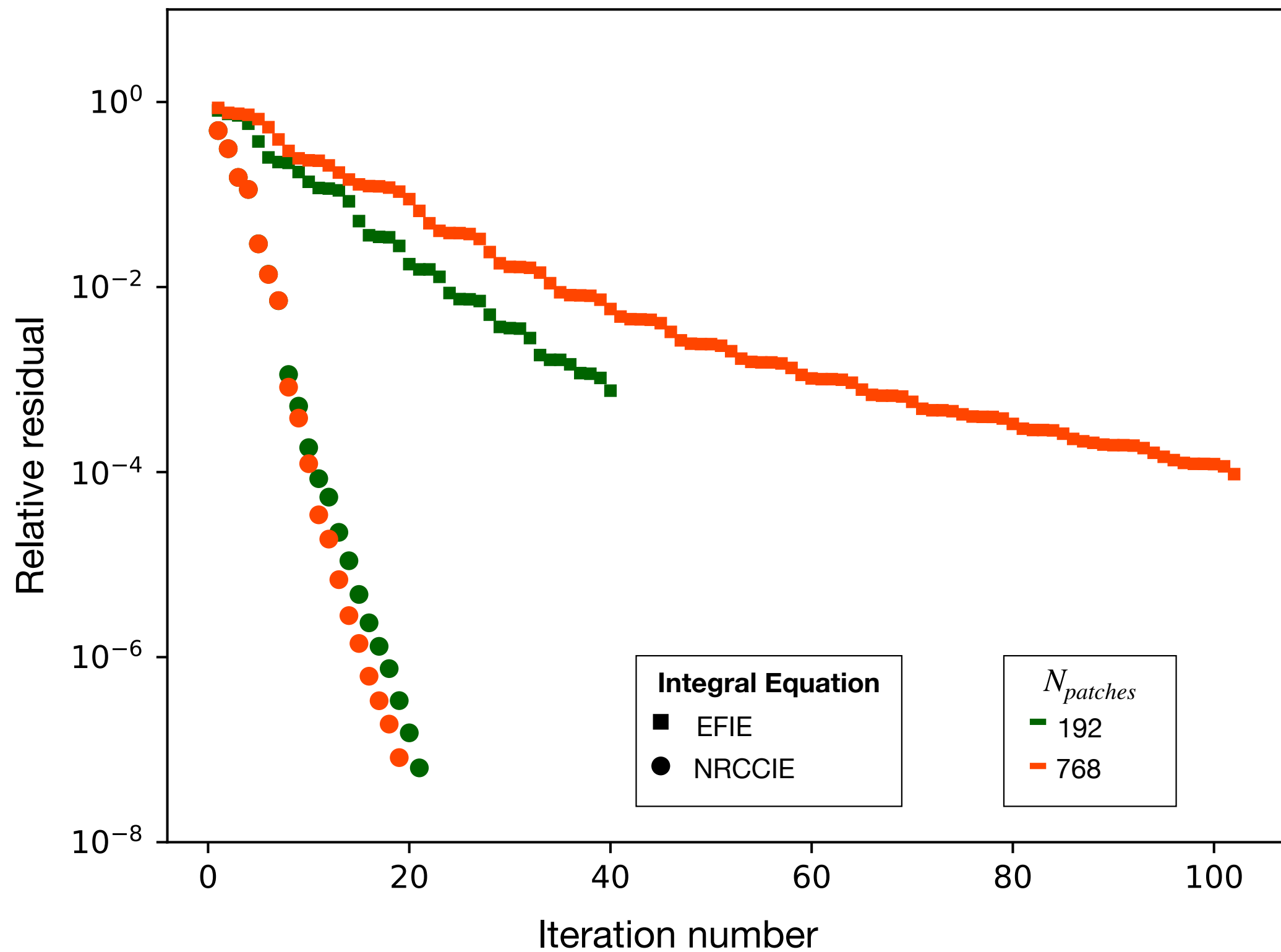
Convergence



Low frequency breakdown



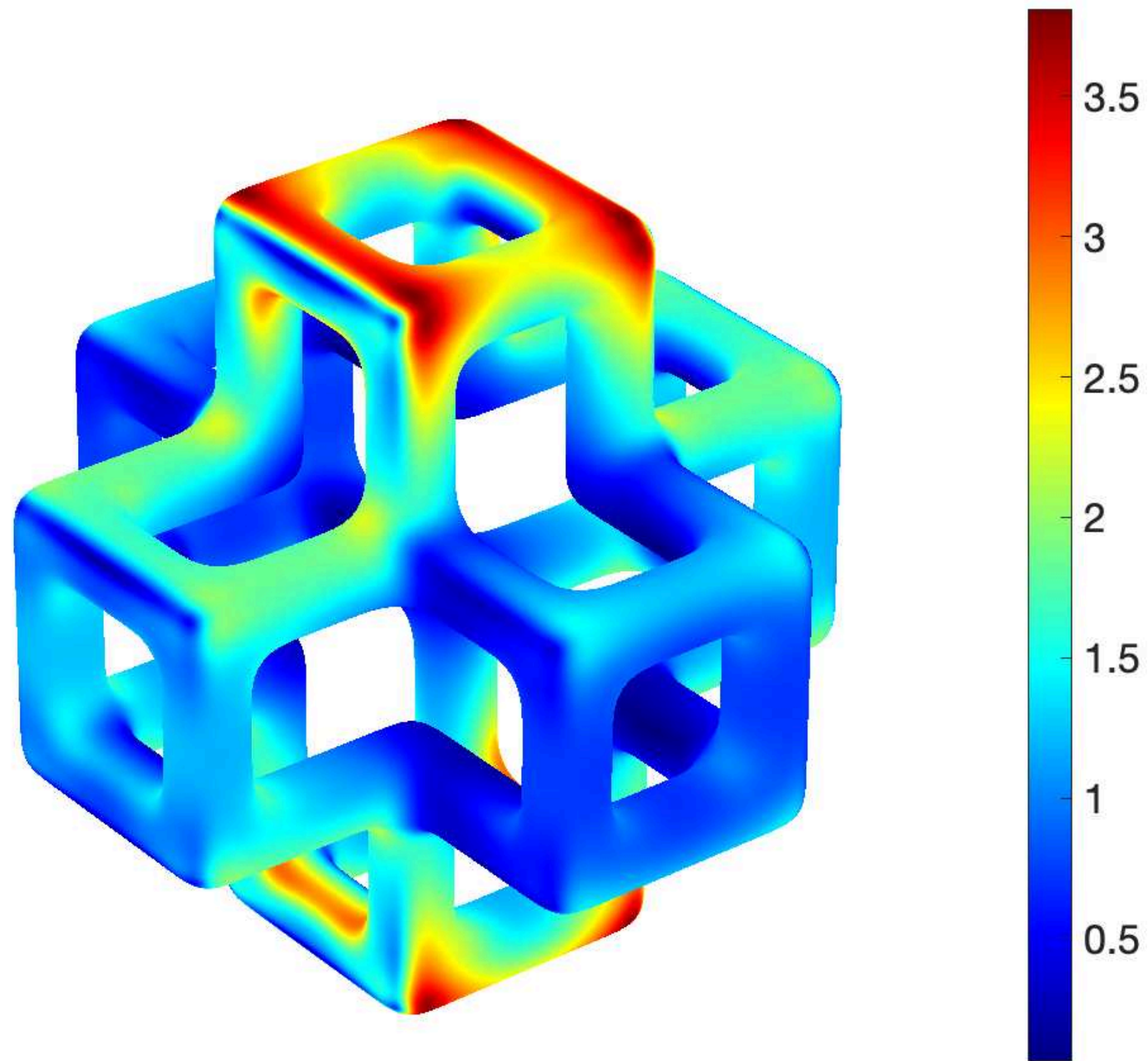
Iterative solver performance



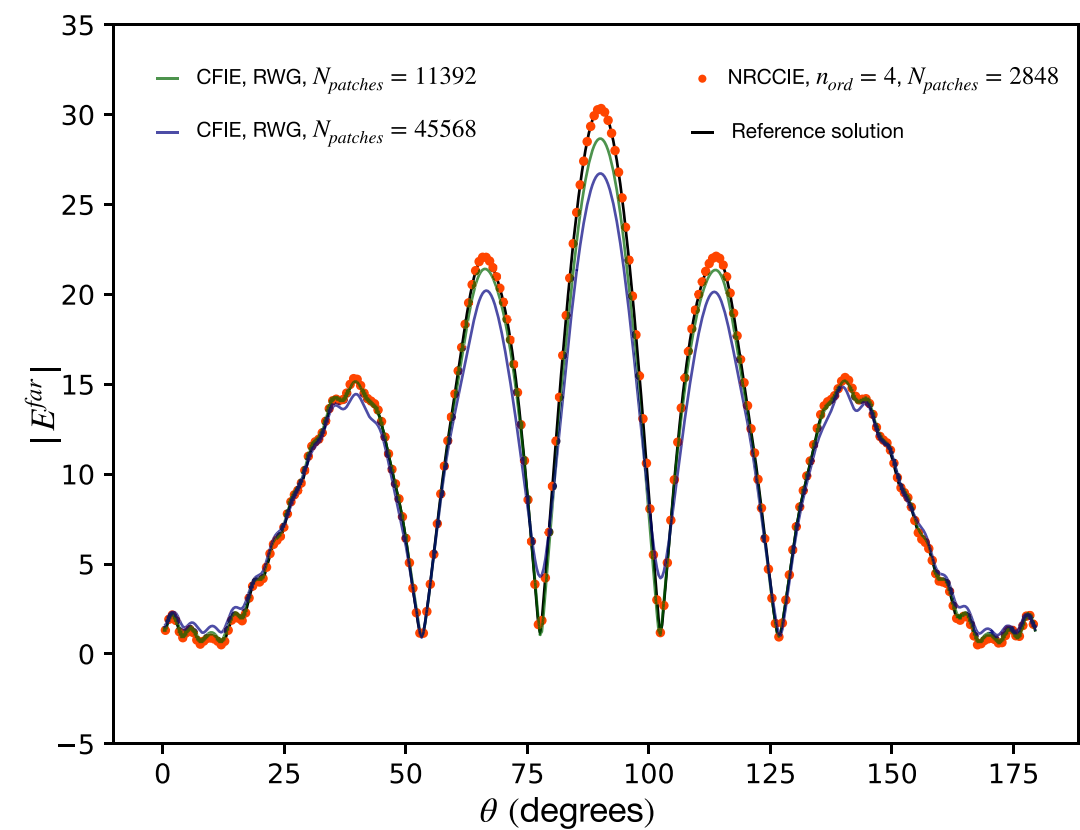
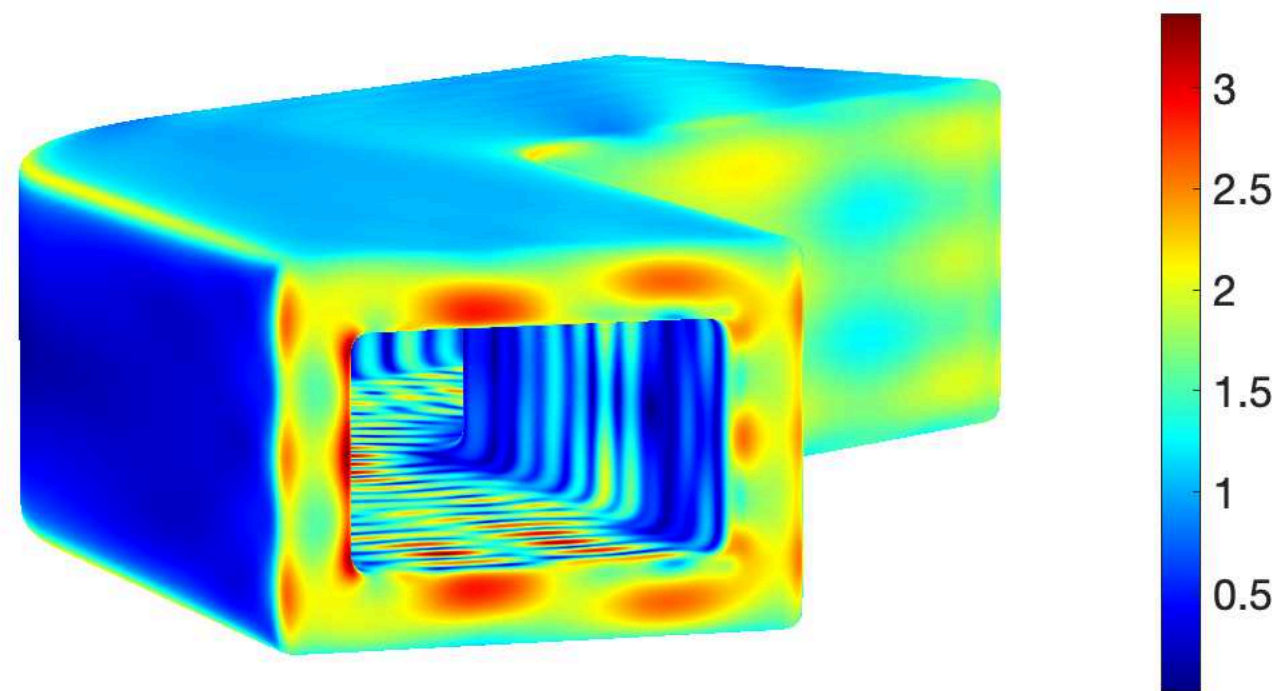
Speed

	n_{ord}	ϵ_{GMRES}	N_{tri}			
			192	768	3072	12288
NRCCIE	2	10^{-7}	2.35	7.16	42.6	153
	4	10^{-9}	14.8	56.6	274	-
	6	10^{-10}	41.0	186	-	-
	8	10^{-13}	130	608	-	-
auRCSIE	2	10^{-7}	3.98	12.9	70	260
	4	10^{-9}	25.9	99.0	472	-
	6	10^{-10}	71.6	325	-	-
	8	10^{-13}	213	990	-	-
DPIE	2	10^{-7}	3.34	10.9	59.6	214
	4	10^{-9}	20.7	81.1	375	-
	6	10^{-10}	59.6	270	-	-
	8	10^{-13}	191	864	-	-

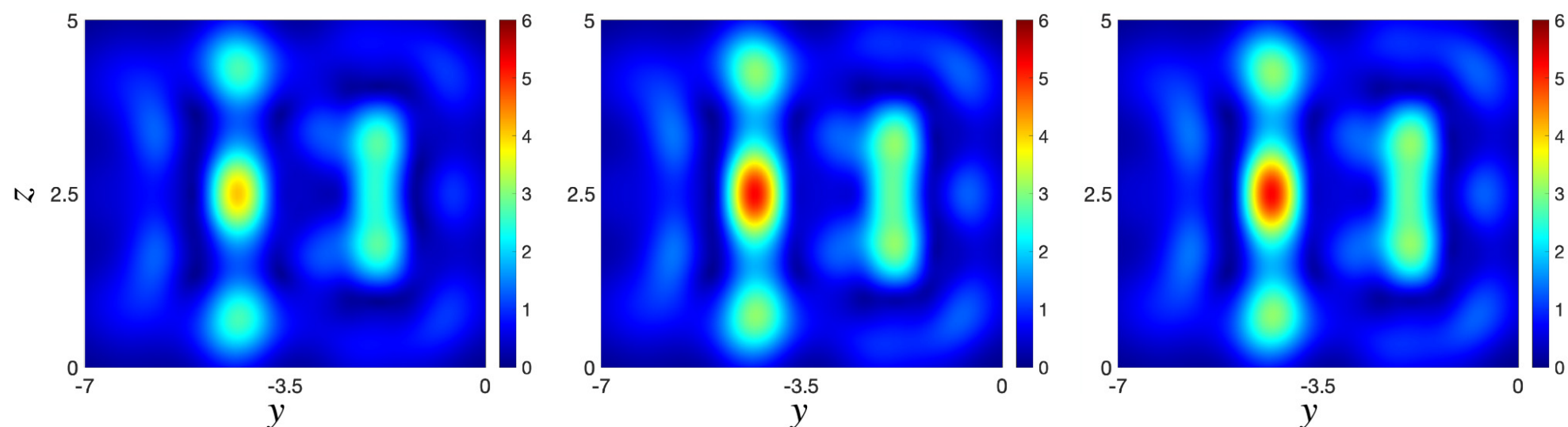
High genus example



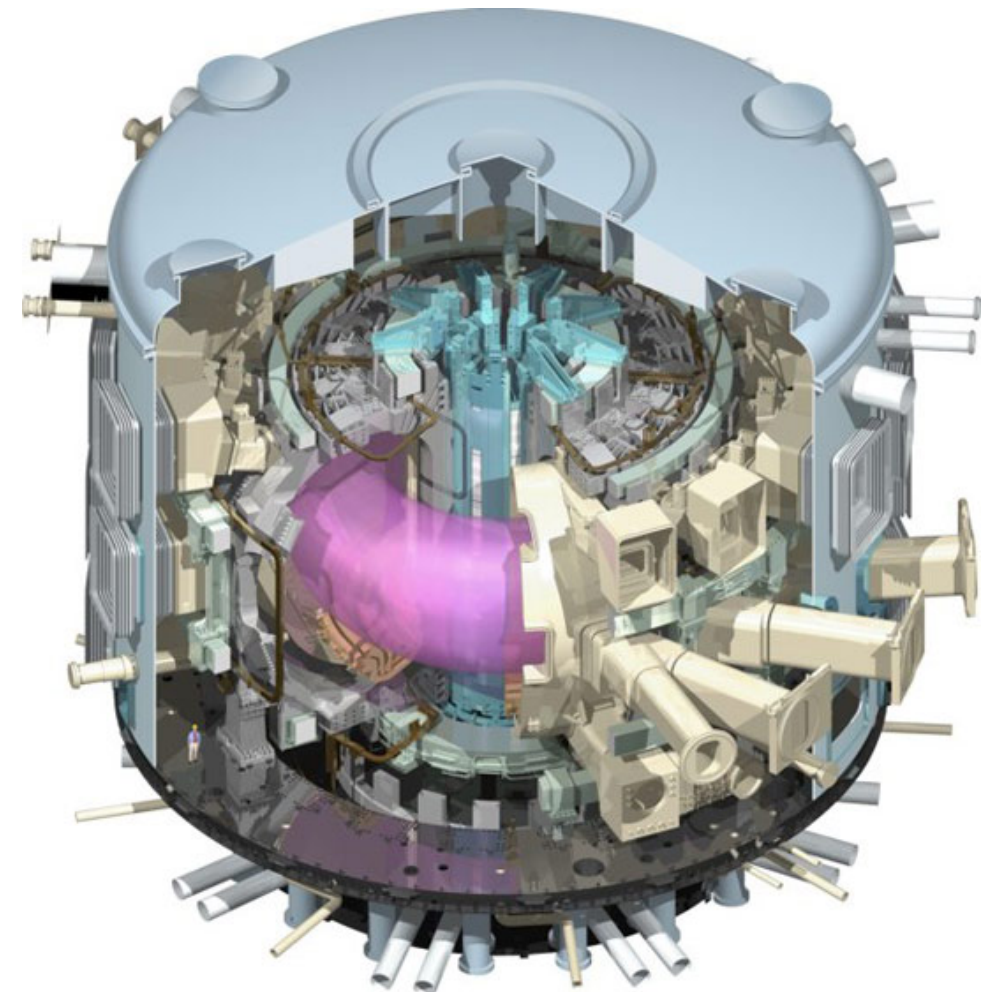
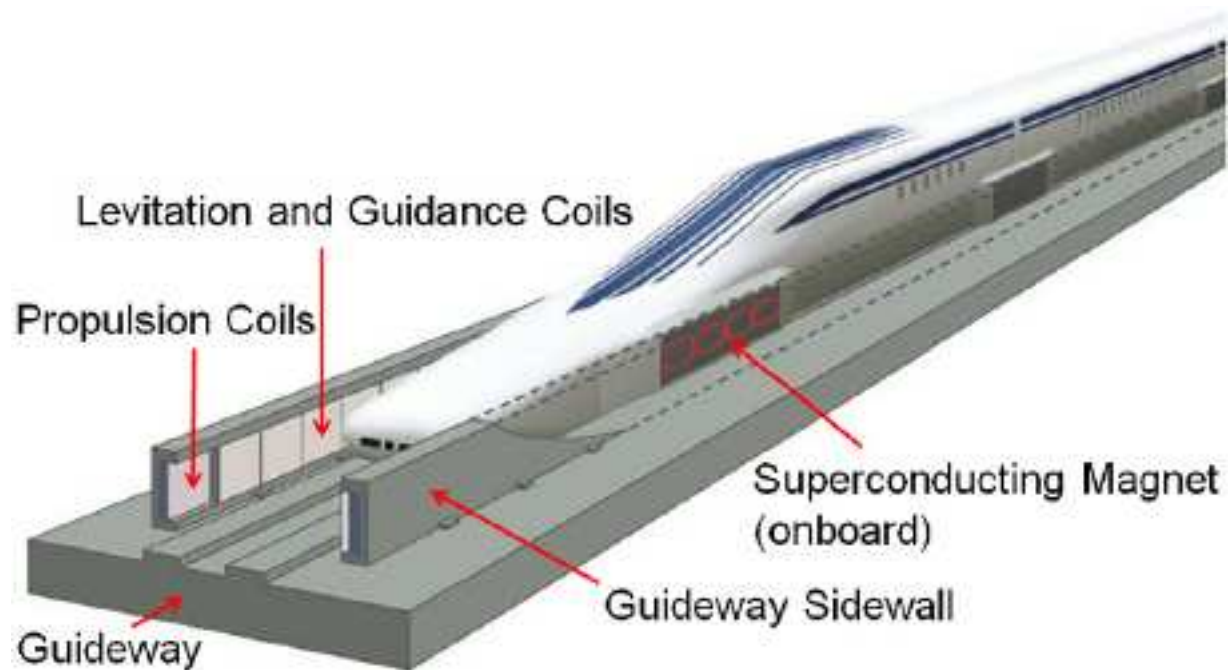
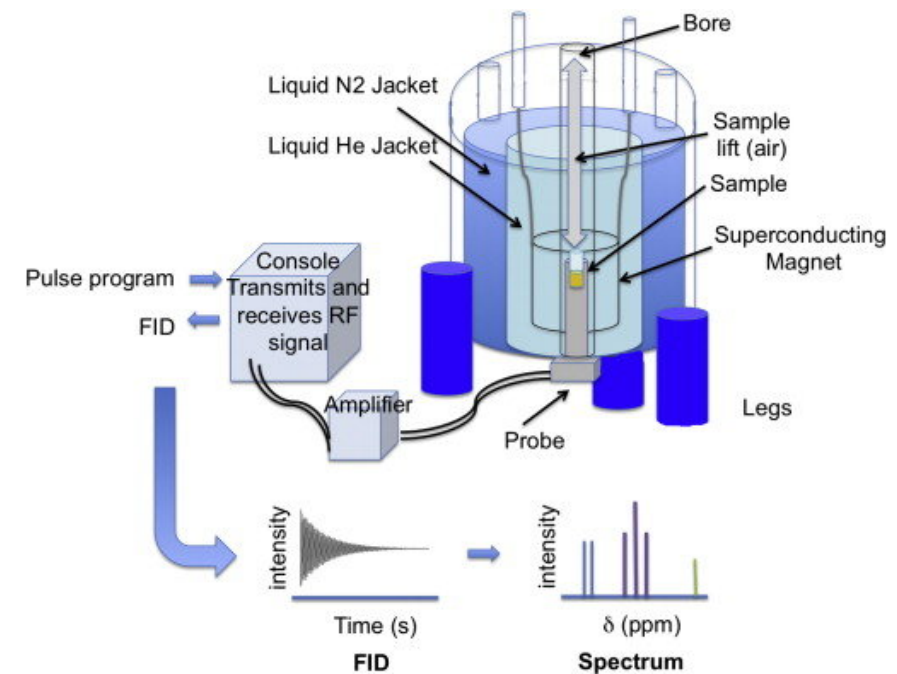
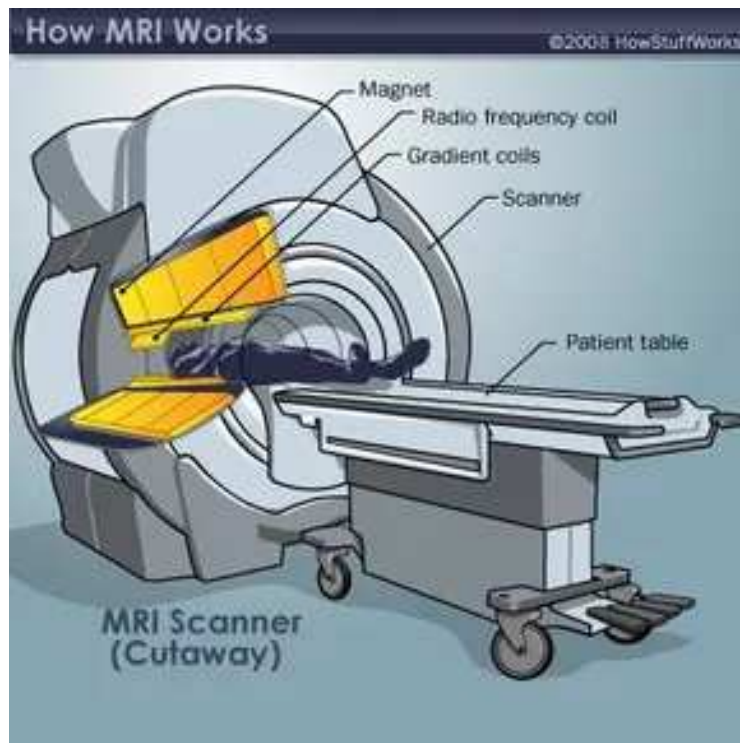
Half-open cavity



	n_{order}	$N_{patches}$	n_{iter}	t_s	ε_f
NRCCIE	2	11392	609	4.5×10^3	6×10^{-3}
	4	2848	607	4.0×10^3	5×10^{-3}
	4	11392	508	7.2×10^3	7×10^{-4}
CFIE	RWG	11392	216	2.5×10^3	1×10^{-1}
	RWG	45568	472	1.7×10^4	5×10^{-2}



Superconductors - an application of generalized Debye formulations



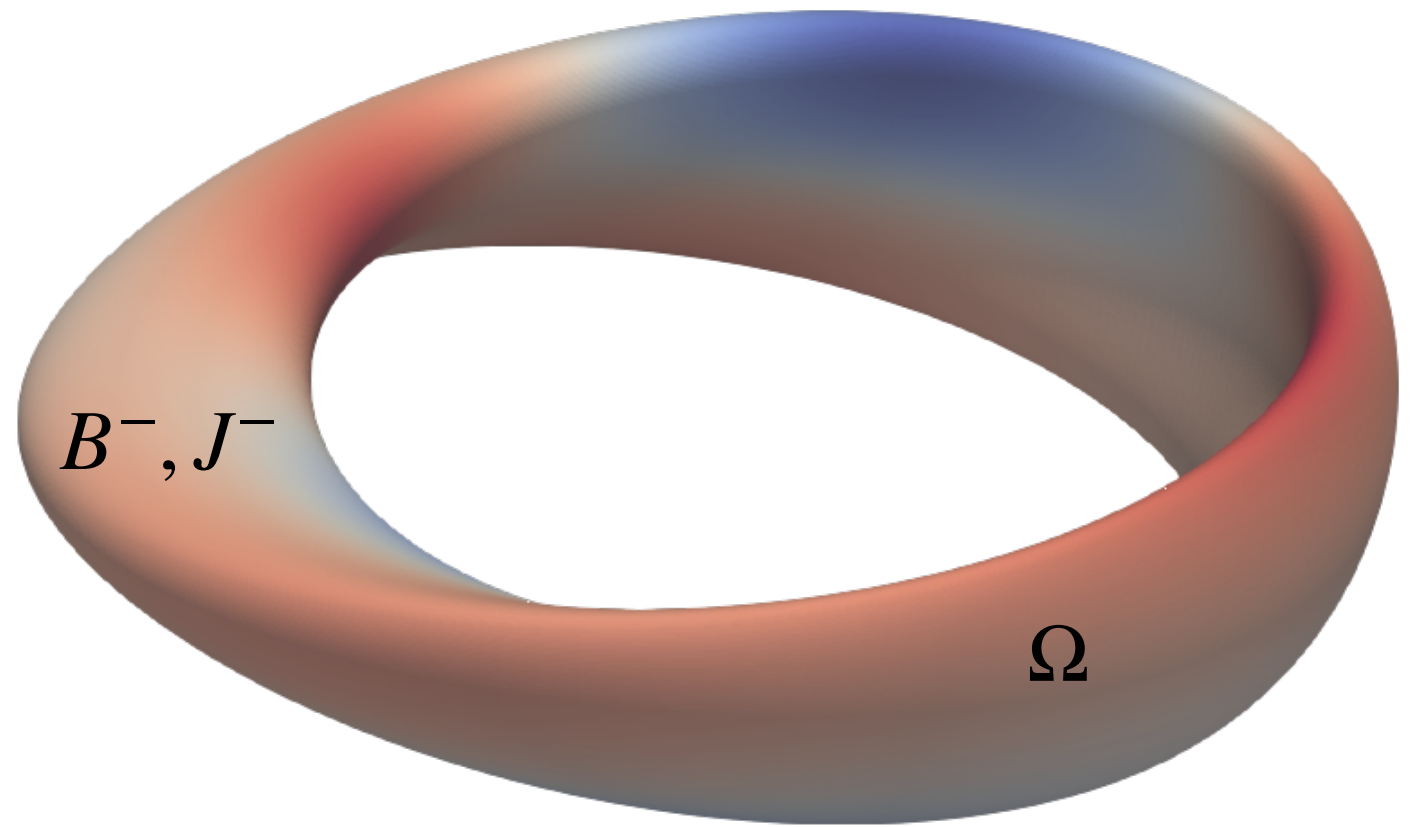
London equations

$$\nabla \times B^- = J^-$$

$$\nabla \times J^- = -\frac{1}{\lambda_L^2} B^- \quad \text{in } \Omega$$

$$\nabla \cdot B^- = 0$$

$$\nabla \cdot J^- = 0$$



London equations

$$\nabla \times B^- = J^-$$

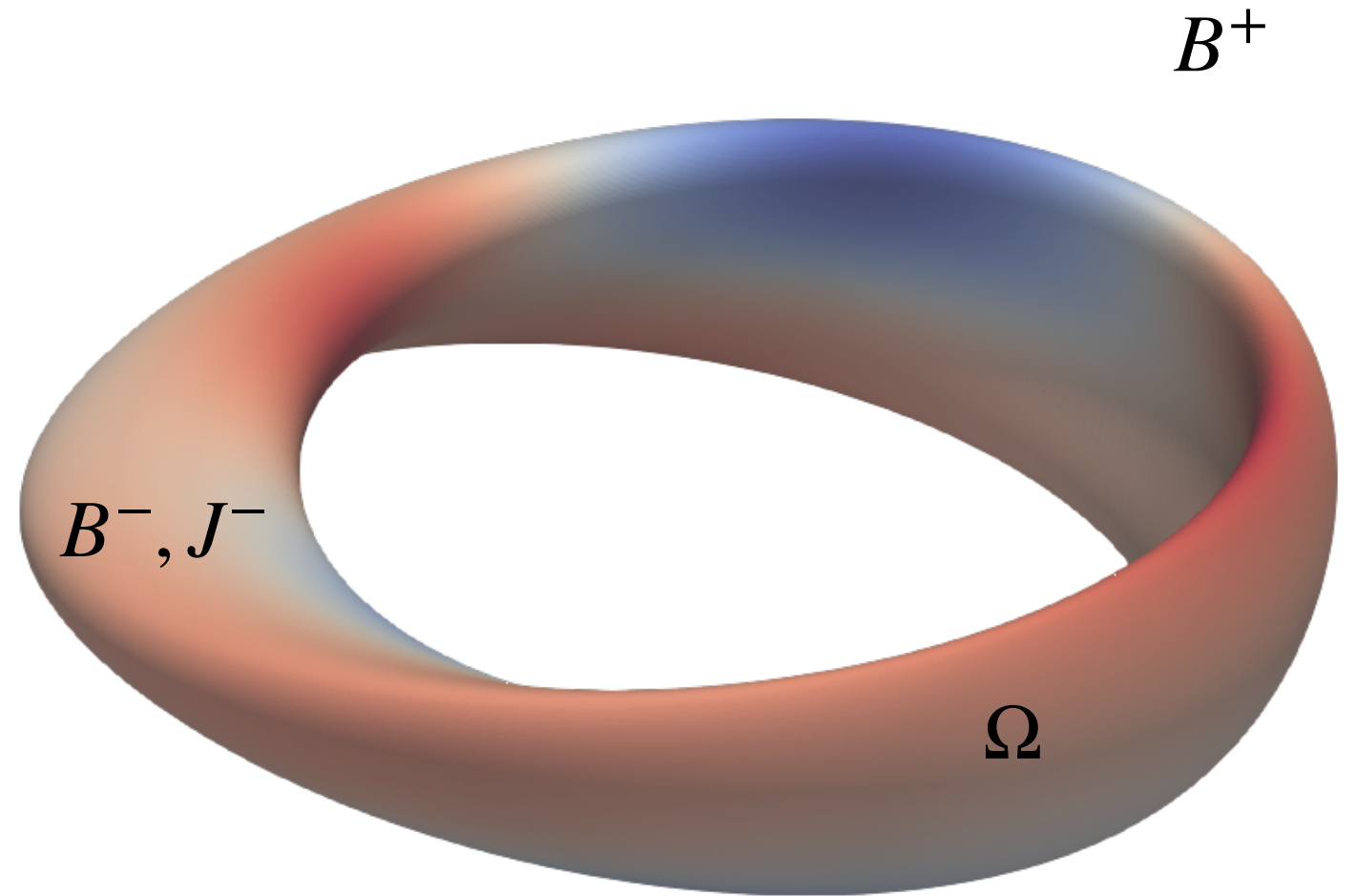
$$\nabla \times J^- = -\frac{1}{\lambda_L^2} B^- \quad \text{in } \Omega$$

$$\nabla \cdot B^- = 0$$

$$\nabla \cdot J^- = 0$$

$$\nabla \times B^+ = 0$$

$$\nabla \cdot B^+ = 0 \quad \text{in } \Omega^c$$



London equations

$$\nabla \times B^- = J^-$$

$$\nabla \times J^- = -\frac{1}{\lambda_L^2} B^- \quad \text{in } \Omega$$

$$\nabla \cdot B^- = 0$$

$$\nabla \cdot J^- = 0$$

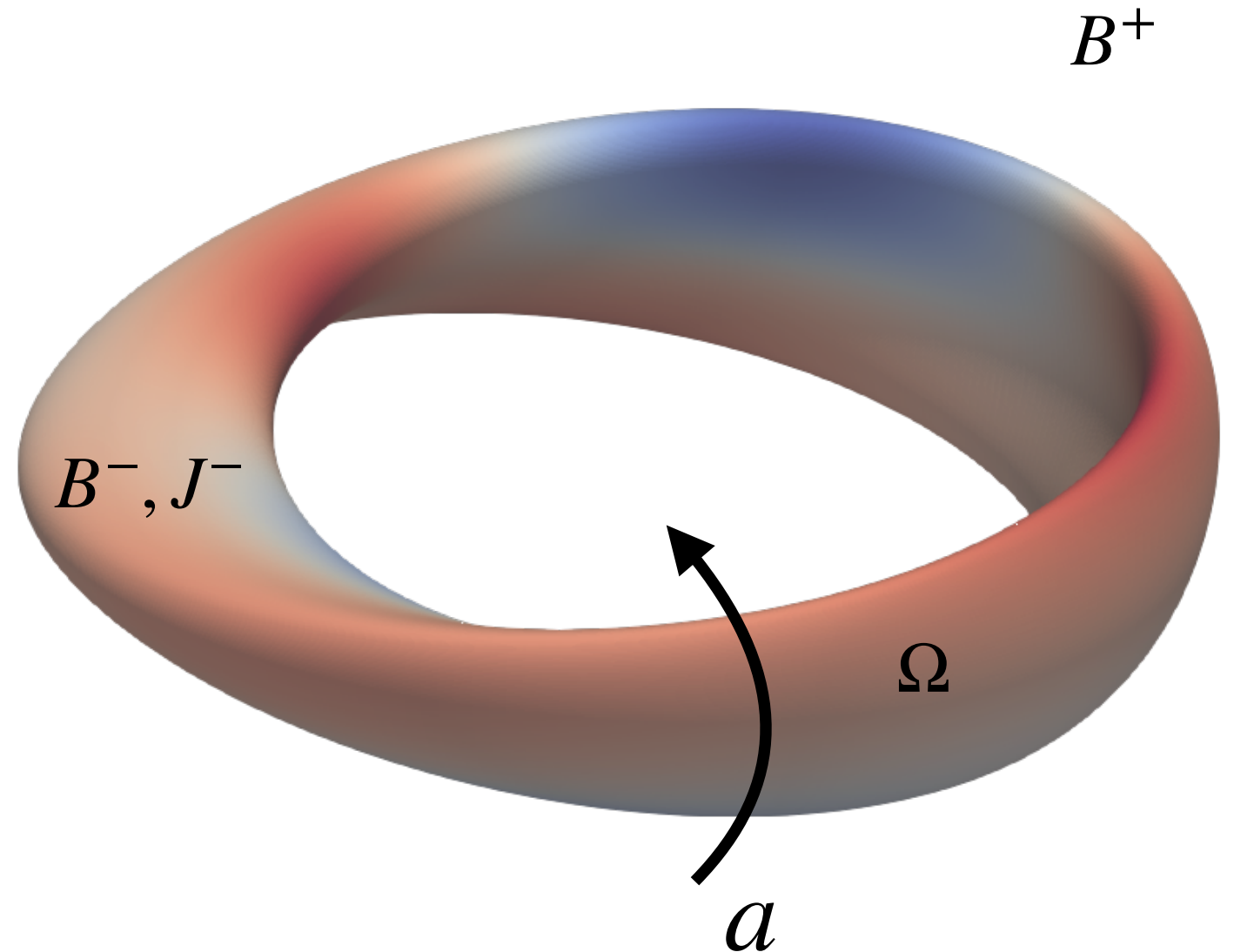
$$\nabla \times B^+ = 0 \quad \text{in } \Omega^c$$

$$\nabla \cdot B^+ = 0$$

$$B^- = B^+ + B^{in} \quad \text{on } \partial\Omega$$

$$\int_{A_j} B^- \cdot d\ell = a_j$$

In case of superconductors $B^{in} \equiv 0$, and the fluxes a_j define the static currents



Uniqueness

$$\nabla \times B^- = J^-$$

$$\nabla \times J^- = -\frac{1}{\lambda_L^2} B^- \quad \text{in } \Omega$$

$$\nabla \cdot B^- = 0$$

$$\nabla \cdot J^- = 0$$

Lorenz gauge



$$B^- = \nabla \times A^-$$

$$J^- = \nabla(\nabla \cdot A^-) - \frac{1}{\lambda_L^2} A^-$$

3 scalar functions,
components of A^-

Uniqueness

$$\nabla \times B^- = J^-$$

$$\nabla \times J^- = -\frac{1}{\lambda_L^2} B^- \quad \text{in } \Omega$$

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Gauge
freedom



$$A^- \rightarrow A^- - \nabla \phi^-$$

3 scalar functions,
components of A^-

Only 2 linearly
independent
scalar functions!

Uniqueness

$$\begin{aligned}\nabla \times B^- &= J^- \\ \nabla \times J^- &= -\frac{1}{\lambda_L^2} B^- \\ \nabla \cdot B^- &= 0 \\ \nabla \cdot J^- &= 0\end{aligned} \quad \text{in } \Omega$$

Lorenz gauge



$$B^- = \nabla \times A^-$$

$$J^- = \nabla(\nabla \cdot A^-) - \frac{1}{\lambda_L^2} A^-$$

Gauge freedom



$$A^- \rightarrow A^- - \nabla \phi^-$$

3 scalar functions,
components of A^-

Only 2 linearly
independent
scalar functions!

$$\begin{aligned}\nabla \times B^+ &= 0 \\ \nabla \cdot B^+ &= 0\end{aligned} \quad \text{in } \Omega^c$$

Curl free



$$B^+ = \nabla f$$

1 scalar function, f

Uniqueness

$$\begin{aligned}\nabla \times B^- &= J^- \\ \nabla \times J^- &= -\frac{1}{\lambda_L^2} B^- \\ \nabla \cdot B^- &= 0 \\ \nabla \cdot J^- &= 0\end{aligned}\quad \text{in } \Omega$$

Lorenz gauge

$$B^- = \nabla \times A^-$$

$$J^- = \nabla(\nabla \cdot A^-) - \frac{1}{\lambda_L^2} A^-$$

Gauge freedom

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3 scalar functions,
components of A^-

Only 2 linearly
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$$\begin{aligned}\nabla \times B^+ &= 0 \\ \nabla \cdot B^+ &= 0\end{aligned}\quad \text{in } \Omega^c$$

Curl free

$$B^+ = \nabla f$$

Non-
contractible
domains

$$B^+ = \nabla f + A_H^+$$

1 scalar function, f

1 scalar function, f +
 g scalar constants

Uniqueness

$$\begin{aligned}\nabla \times B^- &= J^- \\ \nabla \times J^- &= -\frac{1}{\lambda_L^2} B^- \\ \nabla \cdot B^- &= 0 \\ \nabla \cdot J^- &= 0\end{aligned}\quad \text{in } \Omega$$

Lorenz gauge

$$B^- = \nabla \times A^-$$

$$J^- = \nabla(\nabla \cdot A^-) - \frac{1}{\lambda_L^2} A^-$$

Gauge freedom

$$A^- \rightarrow A^- - \nabla \phi^-$$

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Only 2 linearly
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$$\begin{aligned}\nabla \times B^+ &= 0 \\ \nabla \cdot B^+ &= 0\end{aligned}\quad \text{in } \Omega^c$$

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Non-
contractible
domains

$$B^+ = \nabla f + A_H^+$$

1 scalar function, f +
 g scalar constants

$$B^- = B^+ + B^{in} \quad \text{on } \partial\Omega$$

$$\int_{A_j} B^- \cdot d\ell = a_j$$

Uniqueness result

Theorem: If $a_j = 0$, and $B^{in} \equiv 0$, then $B^-, J^- \equiv 0$ in Ω and $B^+ \equiv 0$ in Ω^c

$a_j \equiv 0$ implies $\exists$ a harmonic function f , such that $B^+ = \nabla f$

Integration by parts:

$$\int_{\Omega^c} |B^+|^2 dV = \int_{\Omega^c} |\nabla f|^2 dV = - \int_{\Gamma} f B^+ \cdot n dS$$

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Maxwell integration by parts:

$$\int_{\Omega} \left[\frac{1}{\lambda_L^2} |B^-|^2 + |J^-|^2 \right] dV = \int_{\Gamma} n \times B^- \cdot J^- dS$$

Uniqueness result

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Maxwell integration by parts:

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Continuity of B and surface integration by parts:

$$\begin{aligned} \int_{\Gamma} n \times B^- \cdot J^- dS &= \int_{\Gamma} n \times \nabla f \cdot J^- dS, = \int_{\Gamma} \left[-f \nabla \times J^- \cdot n + \cancel{\nabla \times (f J^-) \cdot n} \right] dS, \\ &= \frac{1}{\lambda_L^2} \int_{\Gamma} f B^+ \cdot n dS. \end{aligned}$$

Uniqueness result

Theorem: If $a_j = 0$, and $B^{in} \equiv 0$, then $B^-, J^- \equiv 0$ in Ω and $B^+ \equiv 0$ in Ω^c

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Integration by parts:

$$\int_{\Omega^c} |B^+|^2 dV = \int_{\Omega^c} |\nabla f|^2 dV = - \int_{\Gamma} f B^+ \cdot n dS$$

$$\int_{\Omega} \left[\frac{1}{\lambda_L^2} |B^-|^2 + |J^-|^2 \right] dV + \frac{1}{\lambda_L^2} \int_{\Omega^c} |B^+|^2 dV = 0$$

Maxwell integration by parts:

$$\int_{\Omega} \left[\frac{1}{\lambda_L^2} |B^-|^2 + |J^-|^2 \right] dV = \int_{\Gamma} n \times B^- \cdot J^- dS$$

Continuity of B and surface integration by parts:

$$\begin{aligned} \int_{\Gamma} n \times B^- \cdot J^- dS &= \int_{\Gamma} n \times \nabla f \cdot J^- dS, = \int_{\Gamma} \left[-f \nabla \times J^- \cdot n + \cancel{\nabla \times (f J^-) \cdot n} \right] dS, \\ &= \frac{1}{\lambda_L^2} \int_{\Gamma} f B^+ \cdot n dS. \end{aligned}$$

London equations: nature of solutions

$$\nabla \times B^- = J^-$$

$$\nabla \times J^- = -\frac{1}{\lambda_L^2} B^-$$

$$\nabla \cdot B^- = 0$$

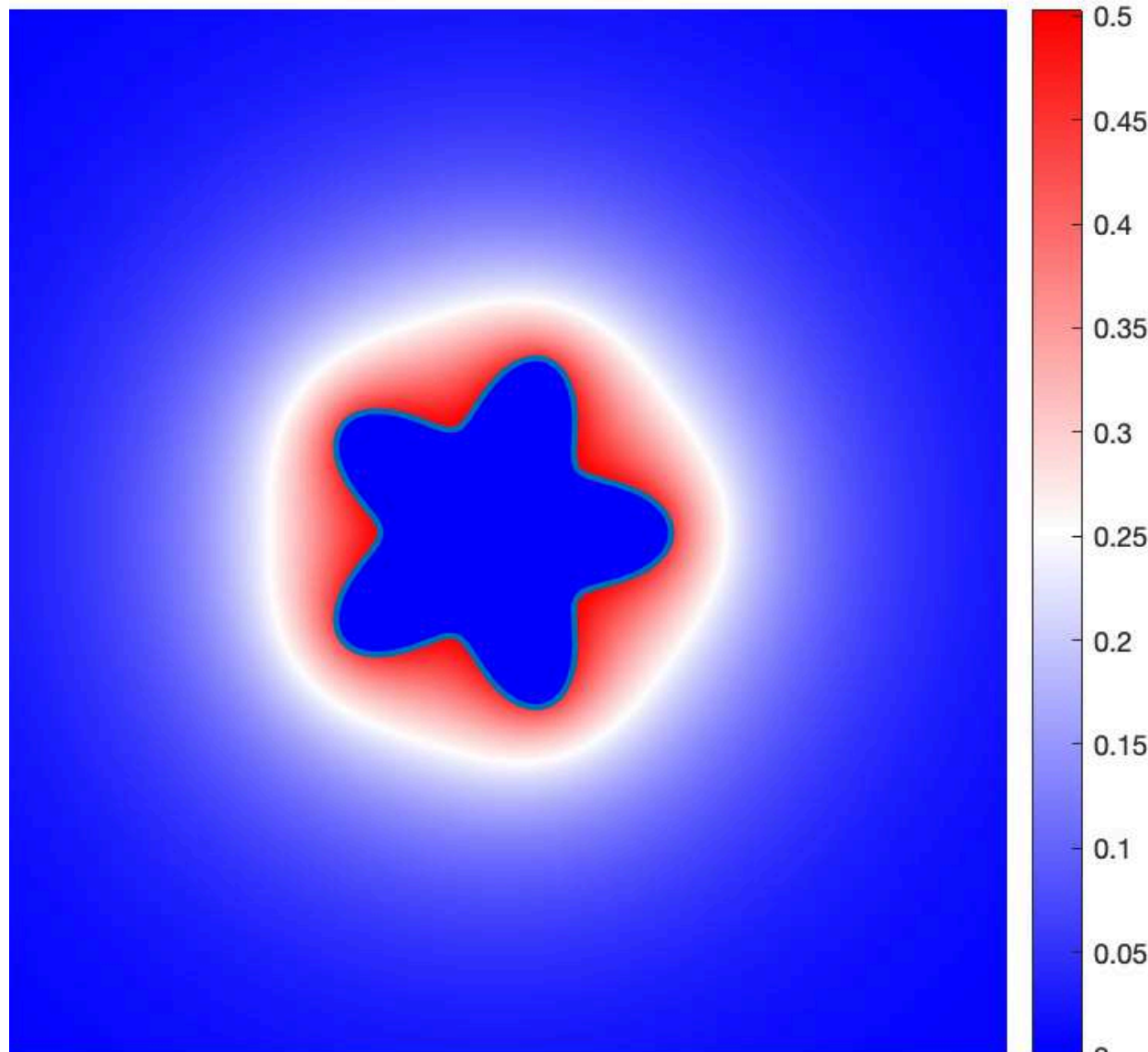
$$\nabla \cdot J^- = 0$$



$$-\Delta B^- + \nabla \nabla \cdot B^- = \nabla \times \nabla \times B^-$$

$$-\Delta B^- + \cancel{\nabla \nabla \cdot B^-} = \nabla \times J^-$$

$$\Delta B^- - \frac{1}{\lambda_L^2} B^- = 0$$



Each component of B^- satisfies
Yukawa equation

$$\lambda_L = 1$$

London equations: nature of solutions

$$\nabla \times B^- = J^-$$

$$\nabla \times J^- = -\frac{1}{\lambda_L^2} B^-$$

$$\nabla \cdot B^- = 0$$

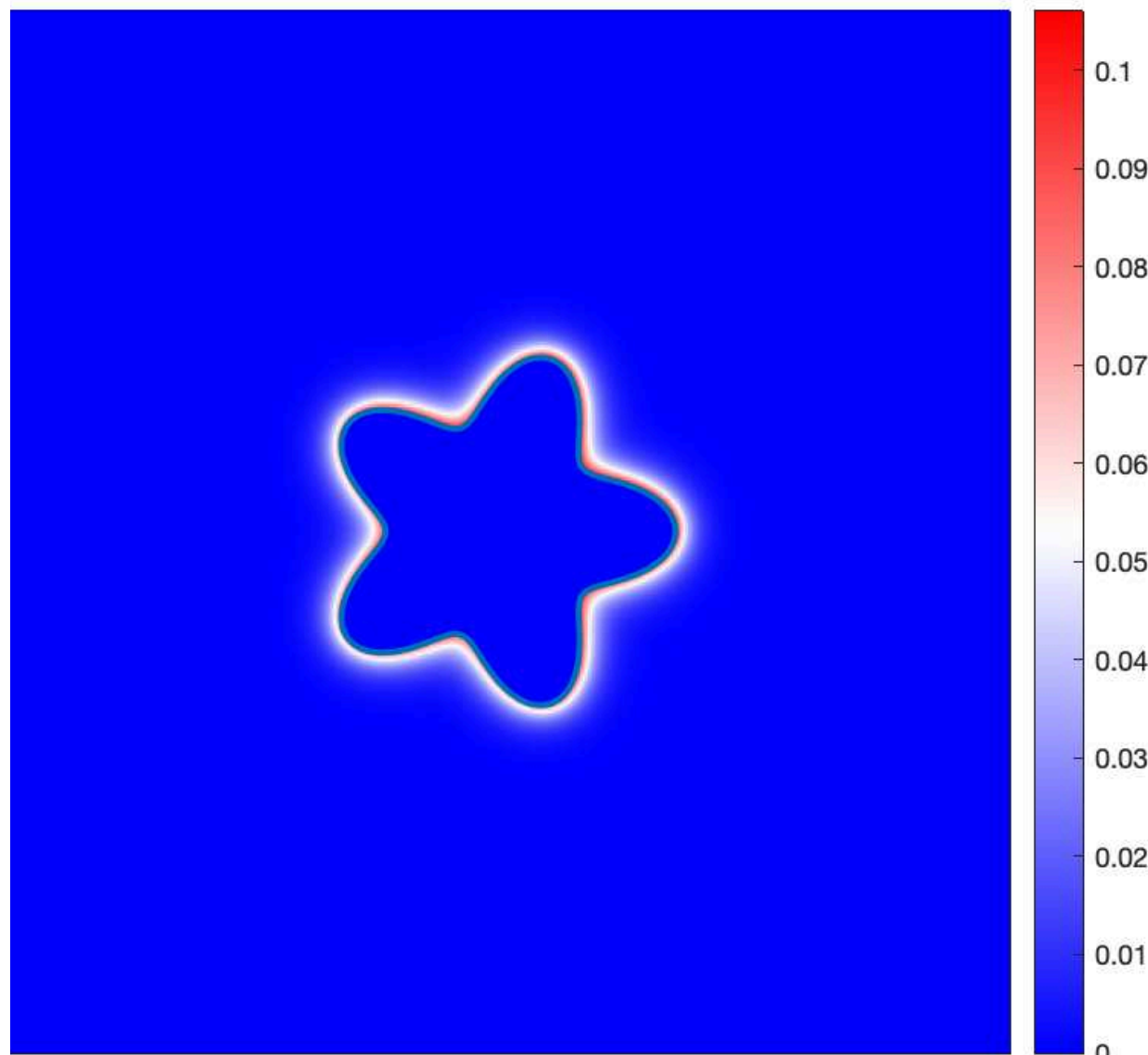
$$\nabla \cdot J^- = 0$$



$$-\Delta B^- + \nabla \nabla \cdot B^- = \nabla \times \nabla \times B^-$$

$$-\Delta B^- + \cancel{\nabla \nabla \cdot B^-} = \nabla \times J^-$$

$$\Delta B^- - \frac{1}{\lambda_L^2} B^- = 0$$



Each component of B^- satisfies
Yukawa equation

$$\lambda_L = 0.1$$

Exterior Representation

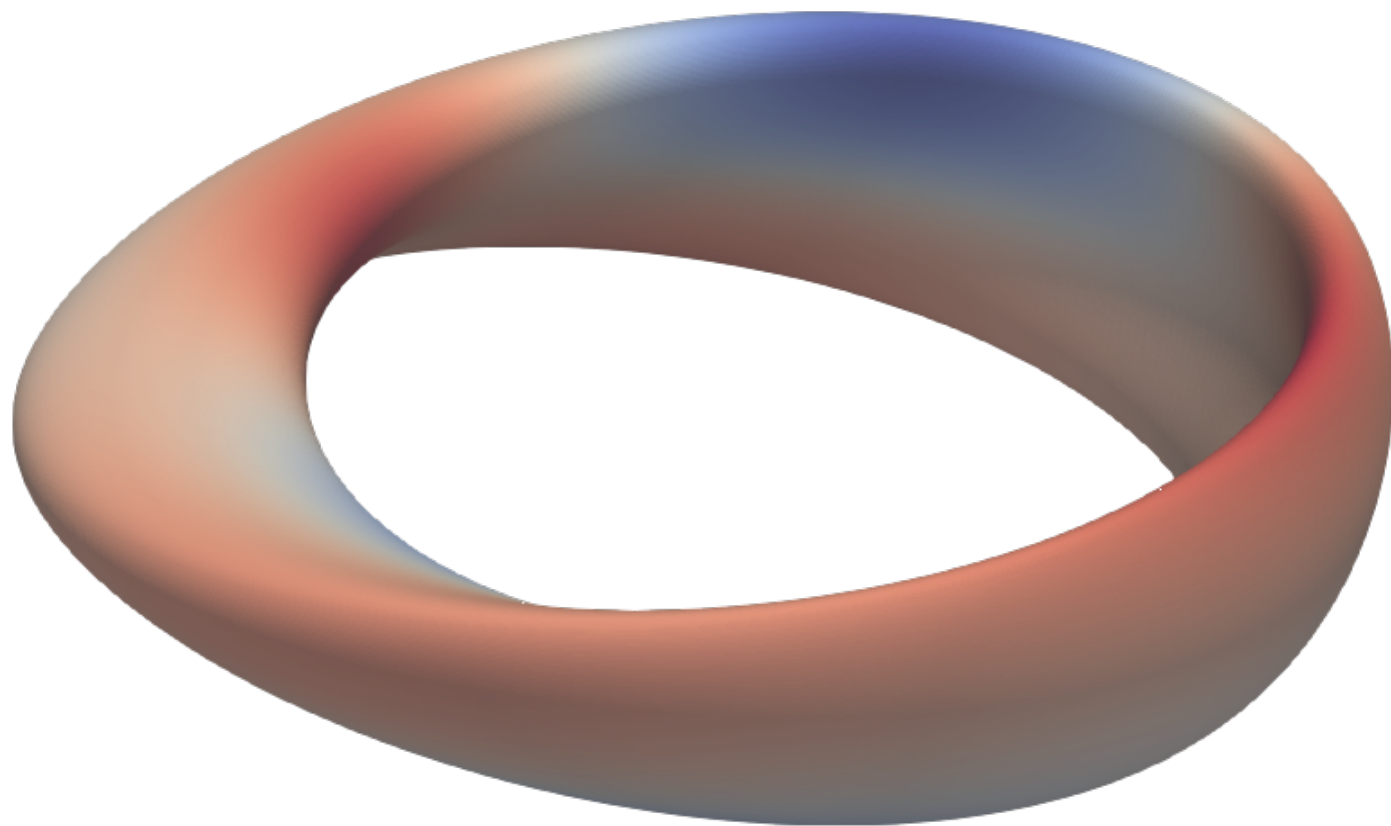
$$\begin{aligned}\nabla \times B^+ &= 0 \\ \nabla \cdot B^+ &= 0\end{aligned}$$

in Ω^c

Curl free



$$B^+ = \nabla f, \text{ where } f \text{ harmonic}$$



Exterior Representation

$$\begin{aligned}\nabla \times B^+ &= 0 \\ \nabla \cdot B^+ &= 0\end{aligned}$$

in Ω^c

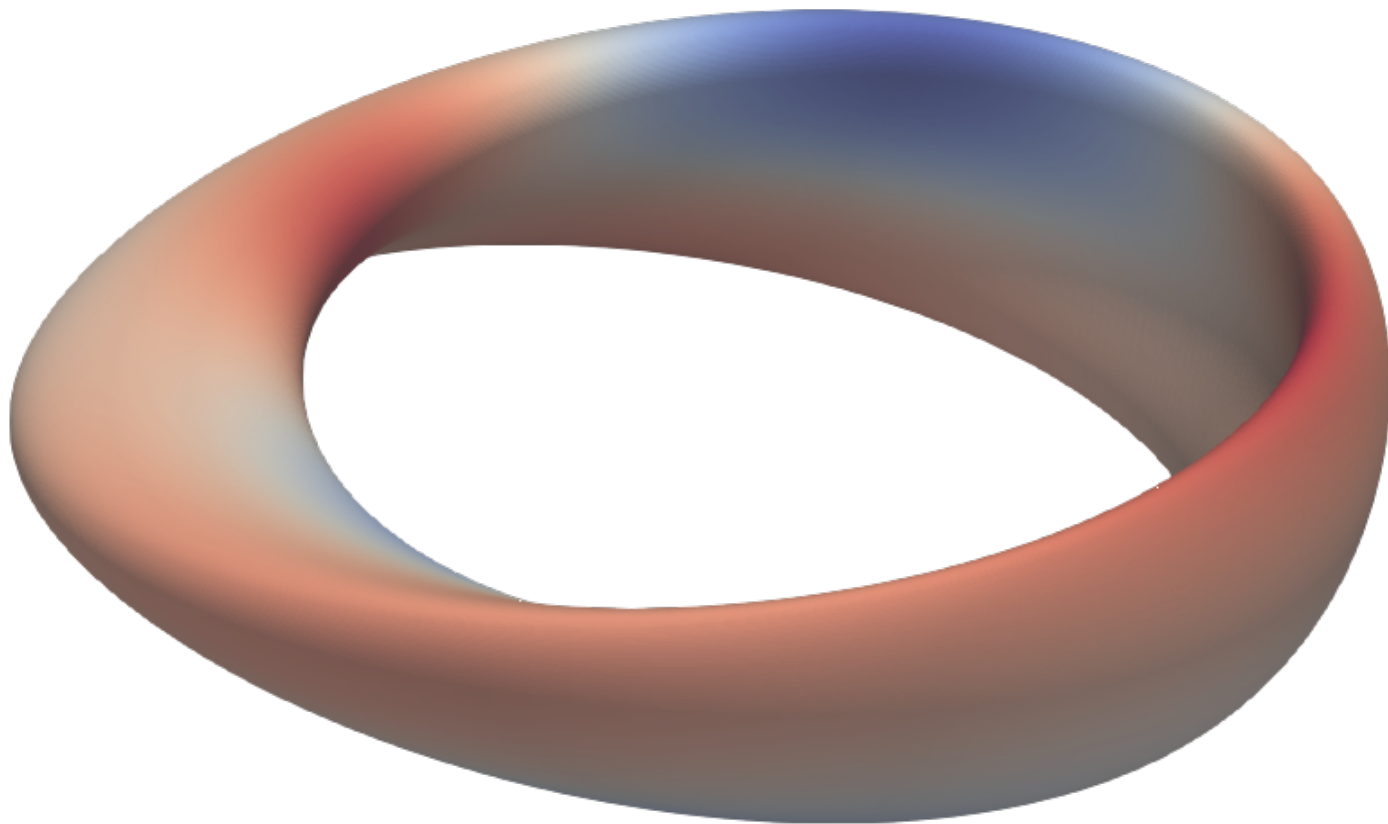
Curl free



$$B^+ = \nabla f, \text{ where } f \text{ harmonic}$$

$$\text{Our choice: } f = \mathcal{S}_0[q^+]$$

$$\mathcal{S}_0[q^+] = \int_{\Gamma} \frac{1}{|x - y|} q^+(y) dS$$



Exterior Representation

$$\begin{aligned}\nabla \times B^+ &= 0 \\ \nabla \cdot B^+ &= 0\end{aligned}$$

in Ω^c

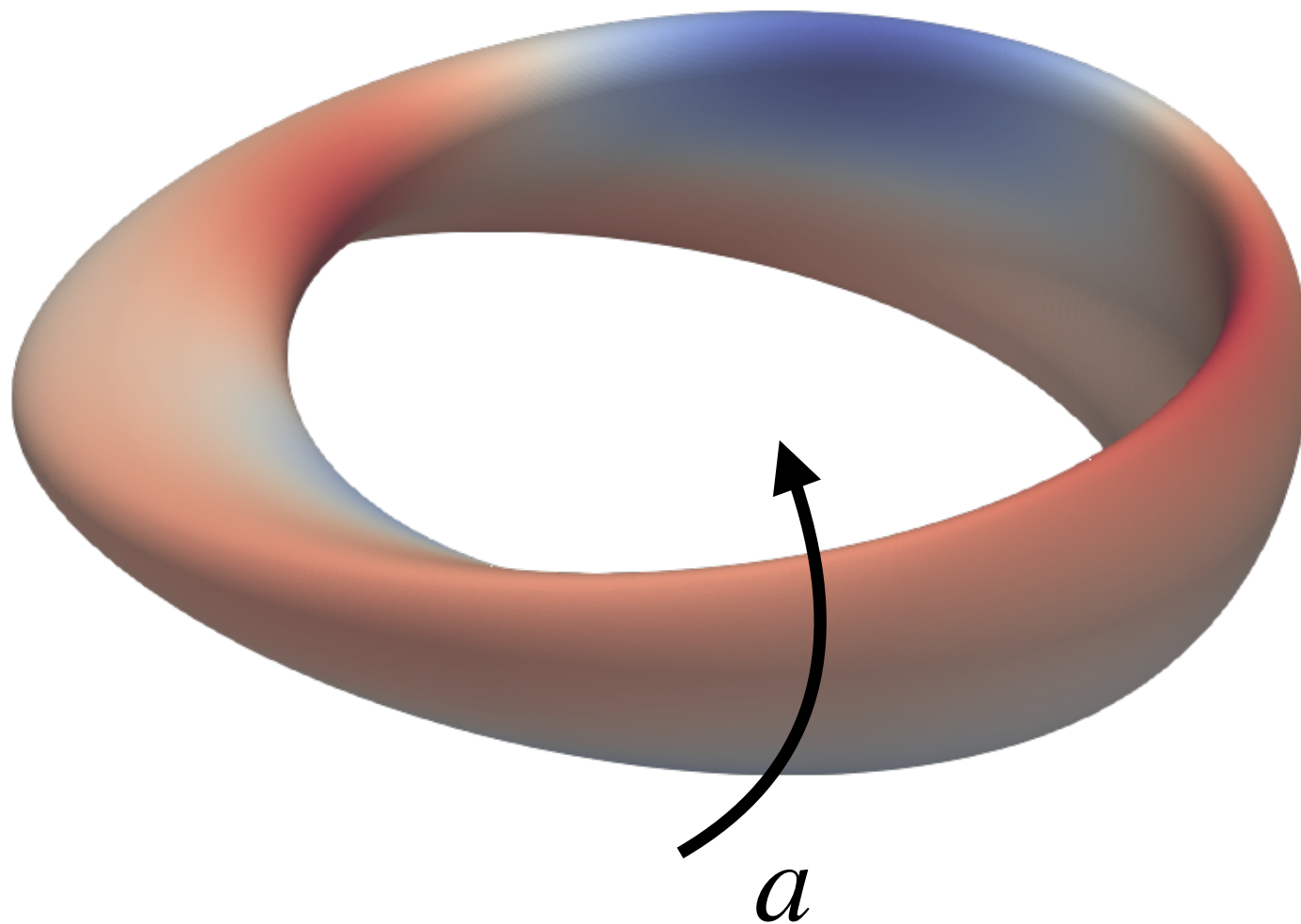
Curl free

$$B^+ = \nabla f, \text{ where } f \text{ harmonic}$$

$$\text{Our choice: } f = \mathcal{S}_0[q^+]$$

$$\mathcal{S}_0[q^+] = \int_{\Gamma} \frac{1}{|x - y|} q^+(y) dS$$

Missing fields due to current loops in 'a' cycles, fix by applying Biot-Savart law to harmonic fields which mimic current loops on 'a' cycles



$$B^+ = \nabla \mathcal{S}_0[q^+] \rightarrow B^+ = \nabla \mathcal{S}_0[q^+] + \nabla \times \mathcal{S}_0[\ell_H^+]$$

ℓ_H^+ : harmonic vector field on Γ

Interior representation

$$\nabla \times B^- = J^-$$

B^-, J^- , a Maxwellian pair

$$\nabla \times J^- = -\frac{1}{\lambda_L^2} B^- \quad \text{in } \Omega$$

$$\nabla \cdot B^- = 0$$

$$\nabla \cdot J^- = 0$$

Lorenz-Debye-Mie, fields can be represented in terms of 2 scalar functions

$$B^- = \nabla \times \nabla \times (ru) + \nabla \times (rv)$$

$$J^- = \nabla \times \nabla \times (rv) - \frac{1}{\lambda_L^2} \nabla \times (ru)$$

where u, v satisfy

$$\left(\Delta - \frac{1}{\lambda_L^2} \right) u = 0, \quad \left(\Delta - \frac{1}{\lambda_L^2} \right) v = 0$$



Interior representation

$$\nabla \times B^- = J^-$$

B^-, J^- , a Maxwellian pair

$$\nabla \times J^- = -\frac{1}{\lambda_L^2} B^- \quad \text{in } \Omega$$

$$\nabla \cdot B^- = 0$$

$$\nabla \cdot J^- = 0$$

Epstein-Greengard: Generalized Debye representation, stable at low frequencies, and results in second kind integral equation

$$B^- = -\mathcal{S}_{\lambda_L}[m^-] + \lambda_L \nabla \mathcal{S}_{\lambda_L}[q^-] + \lambda_L \nabla \times \mathcal{S}_{\lambda_L}[\ell^-]$$

$$J^- = -\frac{1}{\lambda_L} \mathcal{S}_{\lambda_L}[\ell^-] - \nabla \mathcal{S}_{\lambda_L}[r^-] - \nabla \times \mathcal{S}_{\lambda_L}[m^-]$$

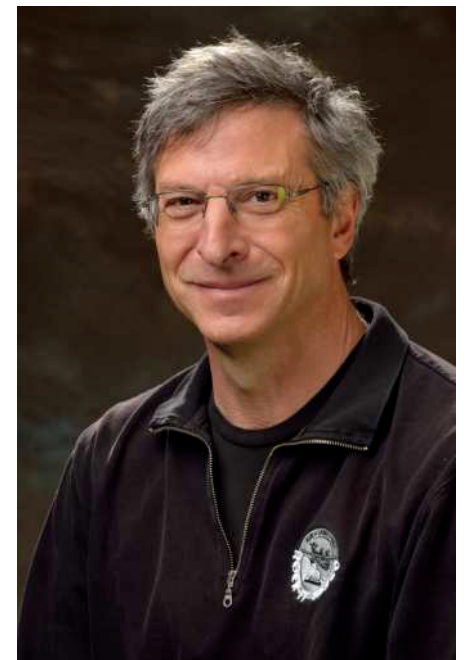
$$\mathcal{S}_{\lambda_L}[q] = \int_{\Gamma} \frac{e^{-\frac{|x-y|}{\lambda_L}}}{|x-y|} q(y) dS$$

$$-\lambda_L \nabla_{\Gamma} \cdot \ell^- = r^-$$

$$\lambda_L \nabla_{\Gamma} \cdot m^- = q^-$$

$$\lambda_L m^- = [\nabla_{\Gamma} \Delta_{\Gamma}^{-1} q^-]$$

$$\lambda_L \ell^- = -[\nabla_{\Gamma} \Delta_{\Gamma}^{-1} r^-] + \ell_H^-$$



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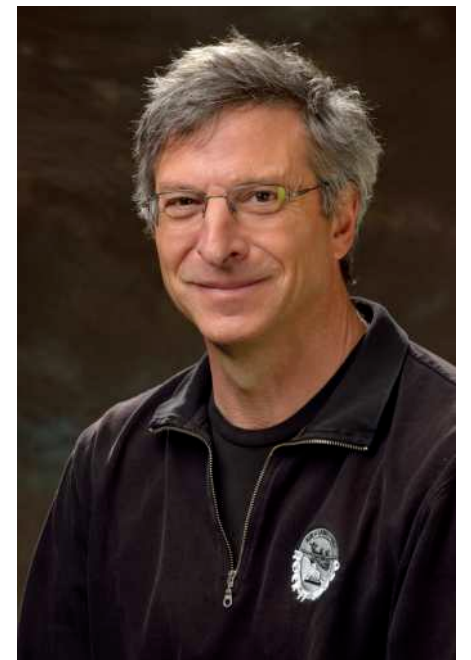
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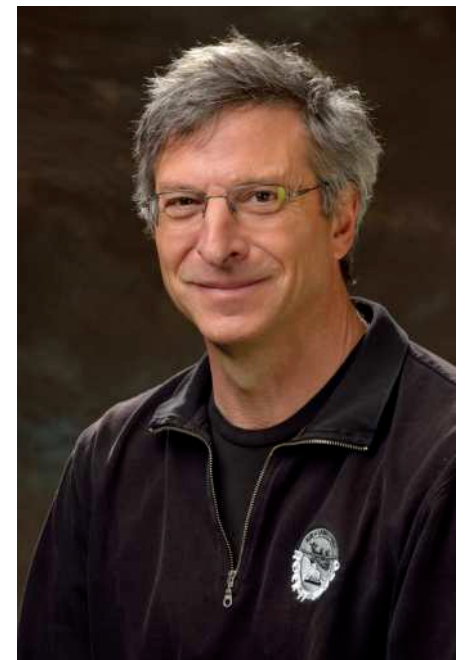
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Representation

$$B^+ = \nabla \mathcal{S}_0[q^+] + \nabla \times \mathcal{S}_0[\ell_H^+]$$

$$\mathcal{S}_0[q^+] = \int_{\Gamma} \frac{1}{|x - y|} q^+(y) dS$$

ℓ_H^+ : harmonic vector field on Γ

Interior representation:

$$B^- = -\mathcal{S}_{\lambda_L}[m^-] + \lambda_L \nabla \mathcal{S}_{\lambda_L}[q^-] + \lambda_L \nabla \times \mathcal{S}_{\lambda_L}[\ell^-]$$

$$J^- = -\frac{1}{\lambda_L} \mathcal{S}_{\lambda_L}[\ell^-] - \nabla \mathcal{S}_{\lambda_L}[r^-] - \nabla \times \mathcal{S}_{\lambda_L}[m^-]$$

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$$\lambda_L \nabla_{\Gamma} \cdot m^- = q^-$$

$$\lambda_L m^- = [\nabla_{\Gamma} \Delta_{\Gamma}^{-1} q^-]$$

$$\lambda_L \ell^- = -[\nabla_{\Gamma} \Delta_{\Gamma}^{-1} r^-] + \ell_H^-$$

ℓ_H^- : harmonic vector field on Γ

$$\mathcal{S}_{\lambda_L}[q] = \int_{\Gamma} \frac{e^{-\frac{|x-y|}{\lambda_L}}}{|x-y|} q(y) dS$$

Unknowns: $q^+, q^-, r^-, \ell_H^-, \ell_H^+$

Integral equation

Unknowns: $q^+, q^-, r^-, \ell_H^-, \ell_H^+$

$$B^+ = \nabla \mathcal{S}_0[q^+] + \nabla \times \mathcal{S}_0[\ell_H^+]$$

ℓ_H^+, ℓ_H^- : harmonic vector fields on Γ

$$B^- = -\mathcal{S}_{\lambda_L}[m^-] + \lambda_L \nabla \mathcal{S}_{\lambda_L}[q^-] + \lambda_L \nabla \times \mathcal{S}_{\lambda_L}[\ell^-]$$

$$\lambda_L m^- = \nabla_\Gamma \Delta_\Gamma^{-1} q^-$$

$$J^- = -\frac{1}{\lambda_L} \mathcal{S}_{\lambda_L}[\ell^-] - \nabla \mathcal{S}_{\lambda_L}[r^-] - \nabla \times \mathcal{S}_{\lambda_L}[m^-]$$

$$\lambda_L \ell^- = -\nabla_\Gamma \Delta_\Gamma^{-1} r^- + \ell_H^-$$

Boundary conditions imposed:

Identity term

$$-\mathcal{S}_0[\nabla_\Gamma \cdot (n \times n \times (B^- - B^+))] = -\mathcal{S}_0[\nabla_\Gamma \cdot (n \times n \times B^{in})]$$

$$(-\lambda_L q^- + q^+)/4$$

$$(B^- - B^+) \cdot n = B^{in} \cdot n$$

$$(\lambda_L q^- + q^+)/2$$

$$\nabla_\Gamma \cdot n \times B^- = n \cdot \nabla \times B^- = J^- \cdot n = \nabla_\Gamma \cdot (n \times B^{in})$$

$$-r^-/2$$

$$\int_{B_j} B^- \cdot d\ell = \int_{B_j} (B^+ + B^{in}) \cdot d\ell = \int_{B_j} B^{in} \cdot d\ell$$

$$\int_{A_j} B^- \cdot d\ell = \int_{A_j} (B^+ + B^{in}) \cdot d\ell = a_j$$

Laplace-beltrami solvers

Computing $\Delta_{\Gamma}^{-1}q$, harmonic vector fields

Suppose $\Delta_{\Gamma}u = q$

Option 1: $u = \mathcal{S}_0^2[\rho]$

$$-\rho/4 + (\mathcal{S}'_0)^2[\rho] - (\mathcal{S}''_0 + \mathcal{D}'_0 - 2H\mathcal{S}'_0)\mathcal{S}_0[\rho] = q$$

Laplace-beltrami solvers

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Option 2: $u(x) = \frac{1}{2\pi} \int_{\Gamma} \log |x - y| \rho(y) dS$

$$\rho + \frac{1}{\pi} \int_{\Gamma} \left(\left(\frac{n(x) \cdot (x - y)}{|x - y|^2} \right)^2 - H(x) \frac{n(x) \cdot (x - y)}{|x - y|^2} \right) dS = q$$

Laplace-beltrami solvers

Computing $\Delta_\Gamma^{-1}q$, harmonic vector fields

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Given a smooth vector tangential field F , on the surface Γ

$$F = \nabla_\Gamma \Delta_\Gamma^{-1} \nabla_\Gamma \cdot F - n \times \nabla_\Gamma \Delta_\Gamma^{-1} \nabla_\Gamma \cdot (n \times F) + H$$

For generic surfaces and smooth V , define $F = n \times n \times V$, and with probability 1, $\text{proj}_H(F) \neq 0$

Static currents



$$\lambda_L = 1$$

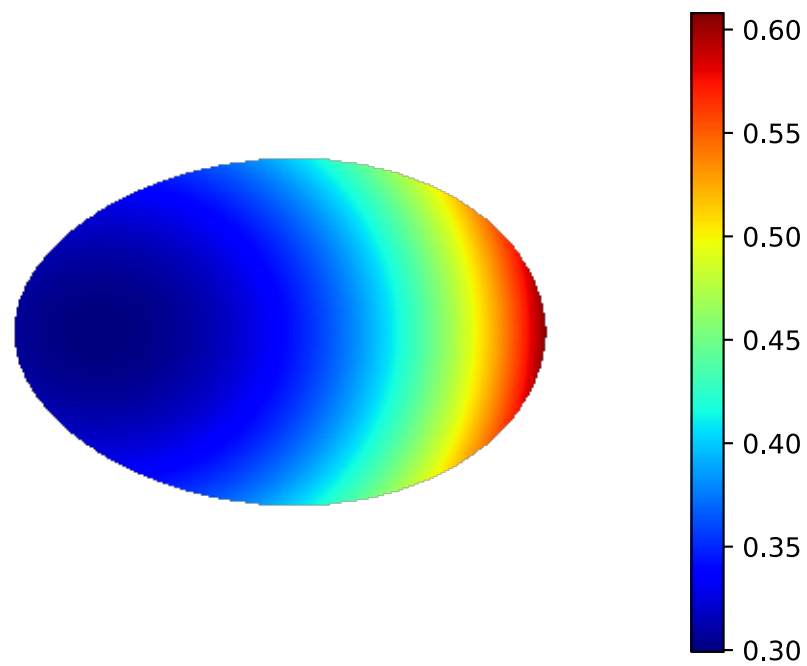


$$\lambda_L = 1/8$$

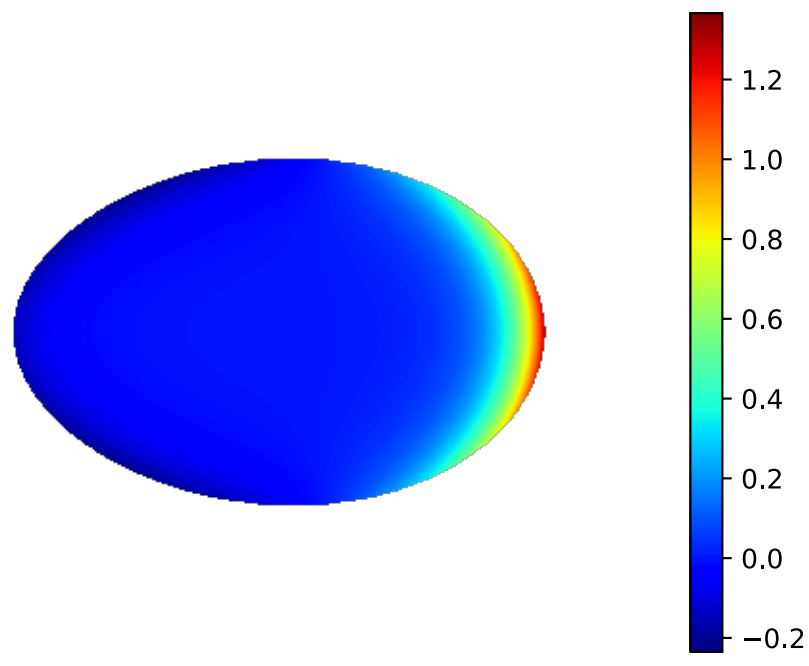


$$\lambda_L = 1/64$$

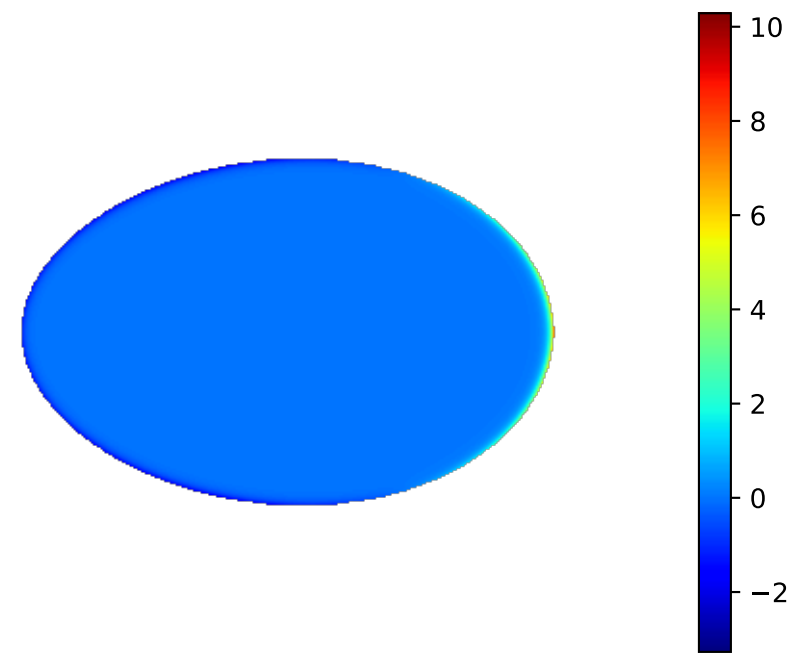
Static currents



$$\lambda_L = 1$$



$$\lambda_L = 1/8$$



$$\lambda_L = 1/64$$

$\lambda_L \rightarrow 0$ - PDE

Decoupled well-conditioned statics problem in the exterior, stabilized by flux conditions, and zero interior fields

$$\nabla \times B_{\lambda_L}^- = J_{\lambda_L}^-$$

$$\nabla \times J_{\lambda_L}^- = -\frac{1}{\lambda_L^2} B_{\lambda_L}^- \quad \text{in } \Omega$$

$$\nabla \cdot B_{\lambda_L}^- = 0$$

$$\nabla \cdot J_{\lambda_L}^- = 0$$

$$\nabla \times B_{\lambda_L}^+ = 0 \quad \text{in } \Omega^c$$

$$\nabla \cdot B_{\lambda_L}^+ = 0$$

$$B_{\lambda_L}^- = B_{\lambda_L}^+ \quad \text{on } \partial\Omega$$

$$\int_{A_j} B_{\lambda_L}^- \cdot d\ell = a_j$$

$$B_0^- = J_0^- \equiv 0 \quad \text{in } \Omega$$

$$\lambda_L \rightarrow 0$$



$$\nabla \times B_0^+ = 0 \quad \text{in } \Omega^c$$

$$\nabla \cdot B_0^+ = 0$$

$$B_0^+ \cdot n = 0 \quad \text{on } \partial\Omega$$

$$\int_{A_j} B_0^+ \cdot d\ell = a_j$$

$\lambda_L \rightarrow 0$ limit - integral equation

$$\begin{array}{lcl}
 B^+ = \nabla \mathcal{S}_0[q^+] + \nabla \times \mathcal{S}_0[\ell_H^+] & & B^+ = \nabla \mathcal{S}_0[q^+] + \nabla \times \mathcal{S}_0[\ell_H^+] \\
 B^- = -\mathcal{S}_{\lambda_L}[m^-] + \lambda_L \nabla \mathcal{S}_{\lambda_L}[q^-] + \lambda_L \nabla \times \mathcal{S}_{\lambda_L}[\ell^-] & \xrightarrow{\lambda_L \rightarrow 0} & B^-|_{\Gamma} = \frac{\nabla_{\Gamma} \Delta_{\Gamma}^{-1} q^- + n \times \ell_H^-}{2} \\
 J^- = -\frac{1}{\lambda_L} \mathcal{S}_{\lambda_L}[\ell^-] - \nabla \mathcal{S}_{\lambda_L}[r^-] - \nabla \times \mathcal{S}_{\lambda_L}[m^-] & & J^- \cdot n|_{\Gamma} = 0 \\
 & & n \times J^-|_{\Gamma} \rightarrow \text{undefined}
 \end{array}$$

Boundary conditions imposed:

$$\begin{array}{lcl}
 -\mathcal{S}_0[\nabla_{\Gamma} \cdot (n \times n \times (B^- - B^+))] = 0 & & B^+ \cdot n = 0 \\
 (B^- - B^+) \cdot n = 0 & \xrightarrow{\lambda_L \rightarrow 0} & \int_{A_j} B^+ \cdot d\ell = a_j \\
 J^- \cdot n = 0 & & \int_{B_j} B^+ \cdot d\ell = 0 \\
 \int_{B_j} B^- \cdot d\ell = \int_{B_j} B^+ \cdot d\ell = 0 & & \\
 \int_{A_j} B^- \cdot d\ell = \int_{A_j} B^+ \cdot d\ell = a_j & &
 \end{array}$$

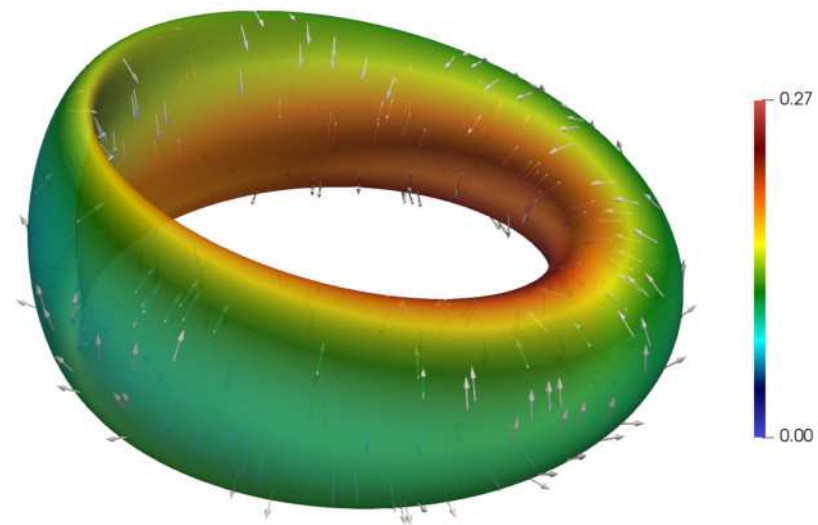
Equations for q^+, ℓ_H^+ decouple completely.

Exterior fields can be directly computed!

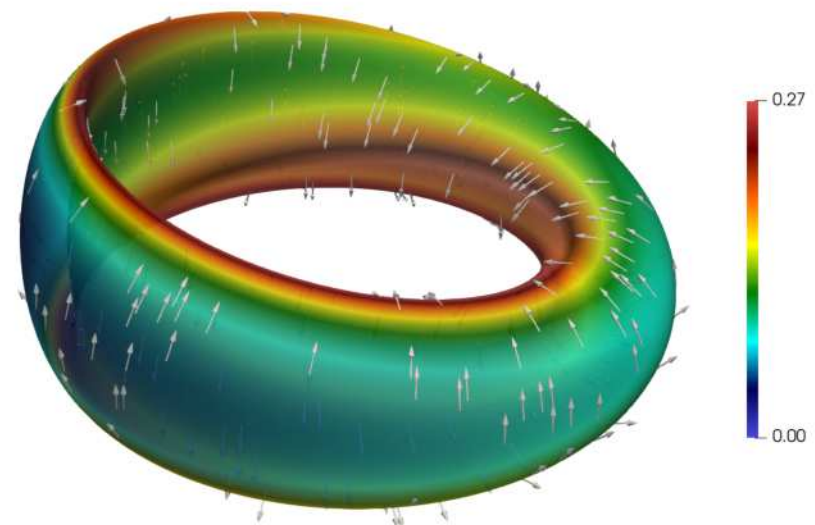
No need to solve for q^-, r^-, ℓ_H^- .

Interior fields on surface read in post-processing

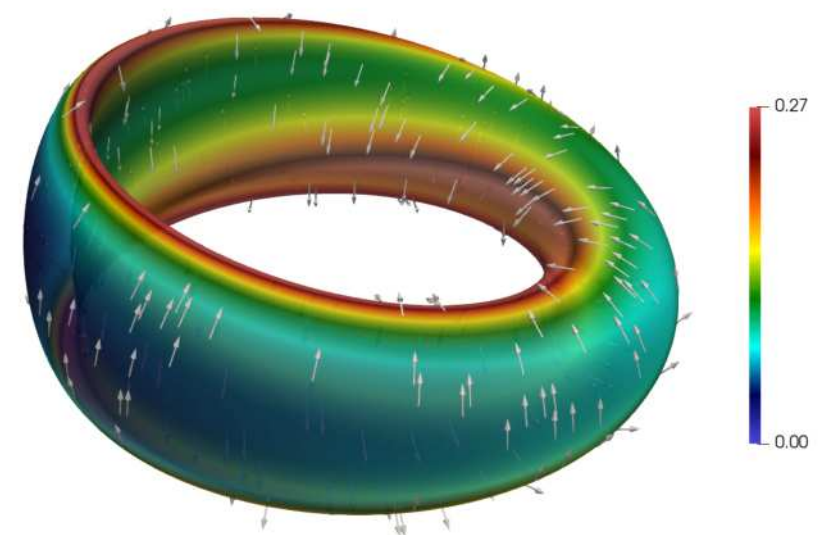
Magnetic fields



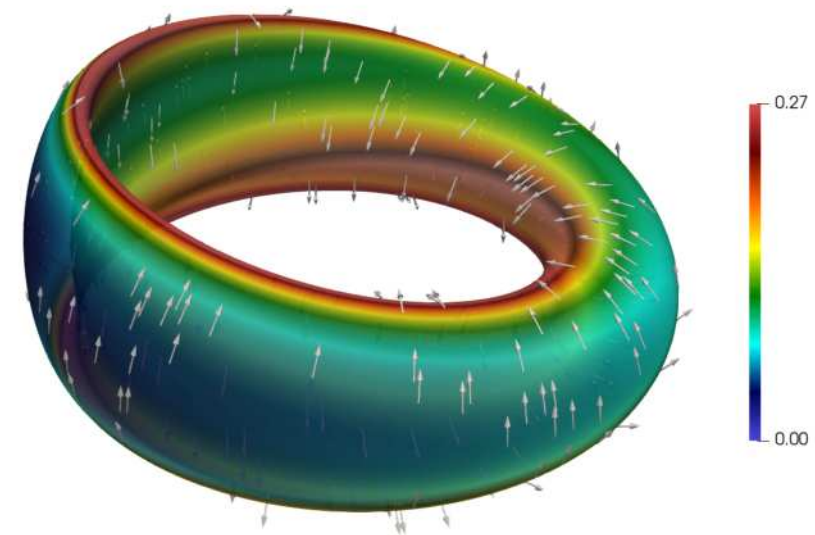
$$\lambda_L = 1$$



$$\lambda_L = 1/8$$

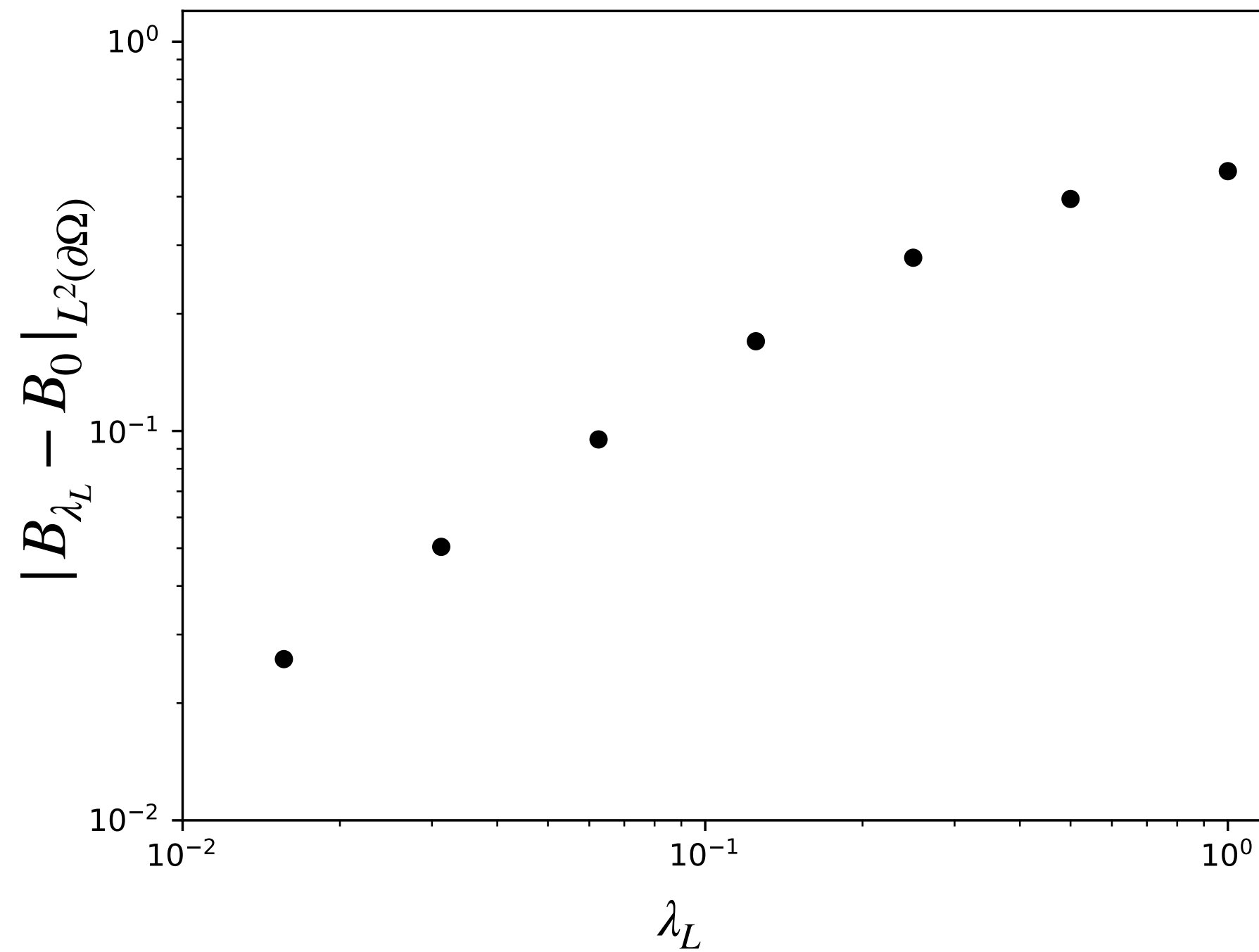


$$\lambda_L = 1/64$$

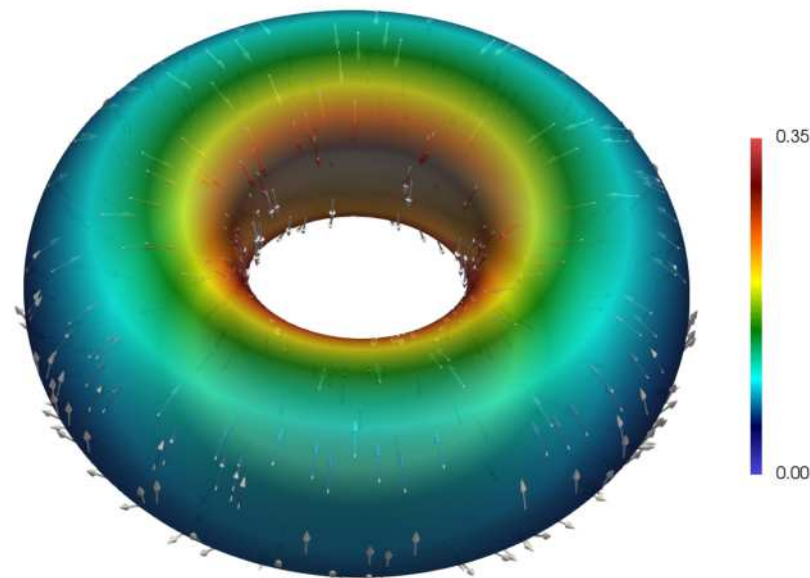


$$\lambda_L = 0$$

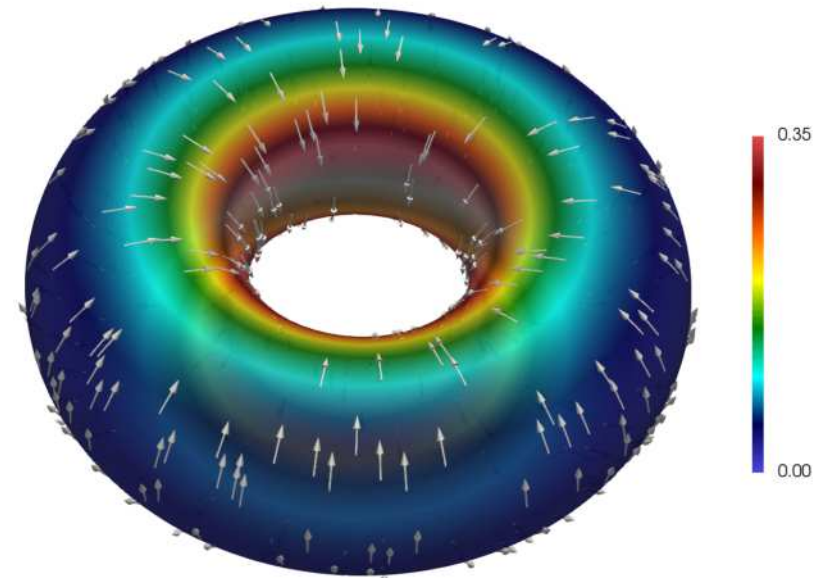
$\lambda_L \rightarrow 0$, numerical results



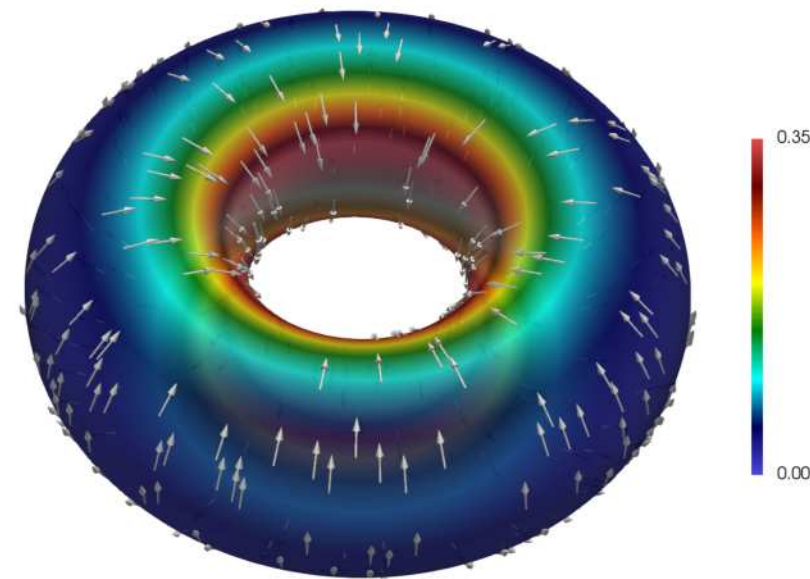
Magnetic fields - multiply connected boundary



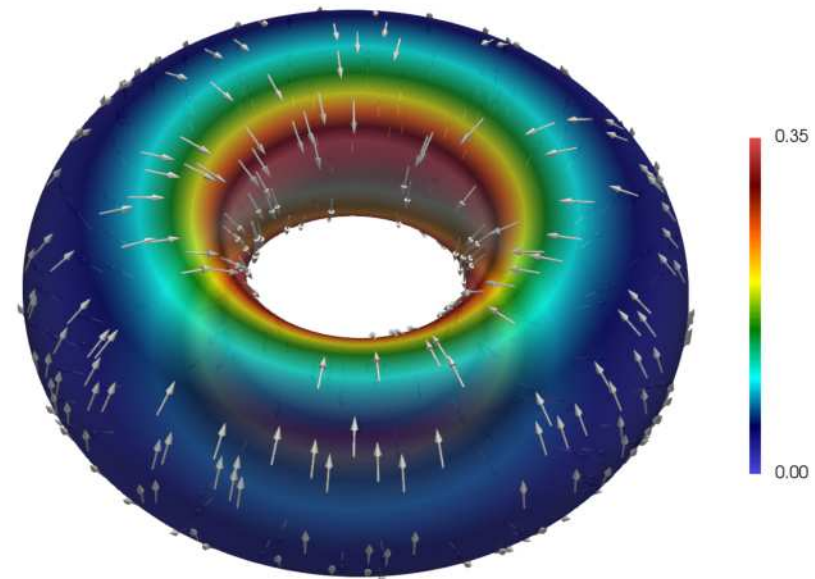
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$$\lambda_L = 1/64$$



$$\lambda_L = 0$$

$\lambda_L \rightarrow 0$, numerical results, multiply connected boundary

